TECHNICAL QUALIFICATION.
PRODUCT INFORMATION.

ASSISTANCE SYSTEMS.



BMW GROUP

General information

Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:



Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as the result of the equipment specification in specific markets or countries.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

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The information contained in this document forms an integral part of the BMW Group Technical Qualification and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

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Technical qualification.

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ASSISTANCE SYSTEMS.
1. INTRODUCTION.

1.1. Introduction

This product information is part of the “BMW Group Automotive Senior Technician" qualification programme and is a reference work for "Assistance systems". The goal of this product information is to provide general background knowledge on the assistance systems used in the current BMW and MINI vehicles.

Target group of qualification programme:

- BMW and MINI employees who have already successfully completed the qualification as “BMW Group Automotive Technician”.
- New BMW and MINI employees who already have basic automotive knowledge and are also familiar with the fundamentals of BMW and MINI vehicles.



These documents make information available that will serve as an overview and give you a better understanding of the qualification content. **As a rule, the instructions and other information in the workshop information system must be used for all maintenance and repair work.**

1.2. Competency prerequisites

There are a number of competencies which are prerequisites for the qualification programme “BMW Group Automotive Senior Technician”. These must either already exist or have been acquired through qualification “BMW Group Automotive Technician”.

In general, the qualification participants should be able to perform general and simple repair work correctly and in accordance with the specifications of the current repair instructions.

In addition, the participants must have the following basic technical knowledge about the assistance systems BMW and MINI vehicles:

Training module	Electrical system I
Extended general vehicle electrical system	Assistance systems

1.3. Assistance systems

Assistance systems help to make the driver's life easier by:

- informing
- suggesting actions
- automatically intervening in the driving process

BMW has always been a pioneer in the development and use of assistance systems. For example, the E32 was the first vehicle in the world to have Park Distance Control.

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The E65 was the first vehicle with a “Night Vision” thermal imaging camera and a high-beam assistant.

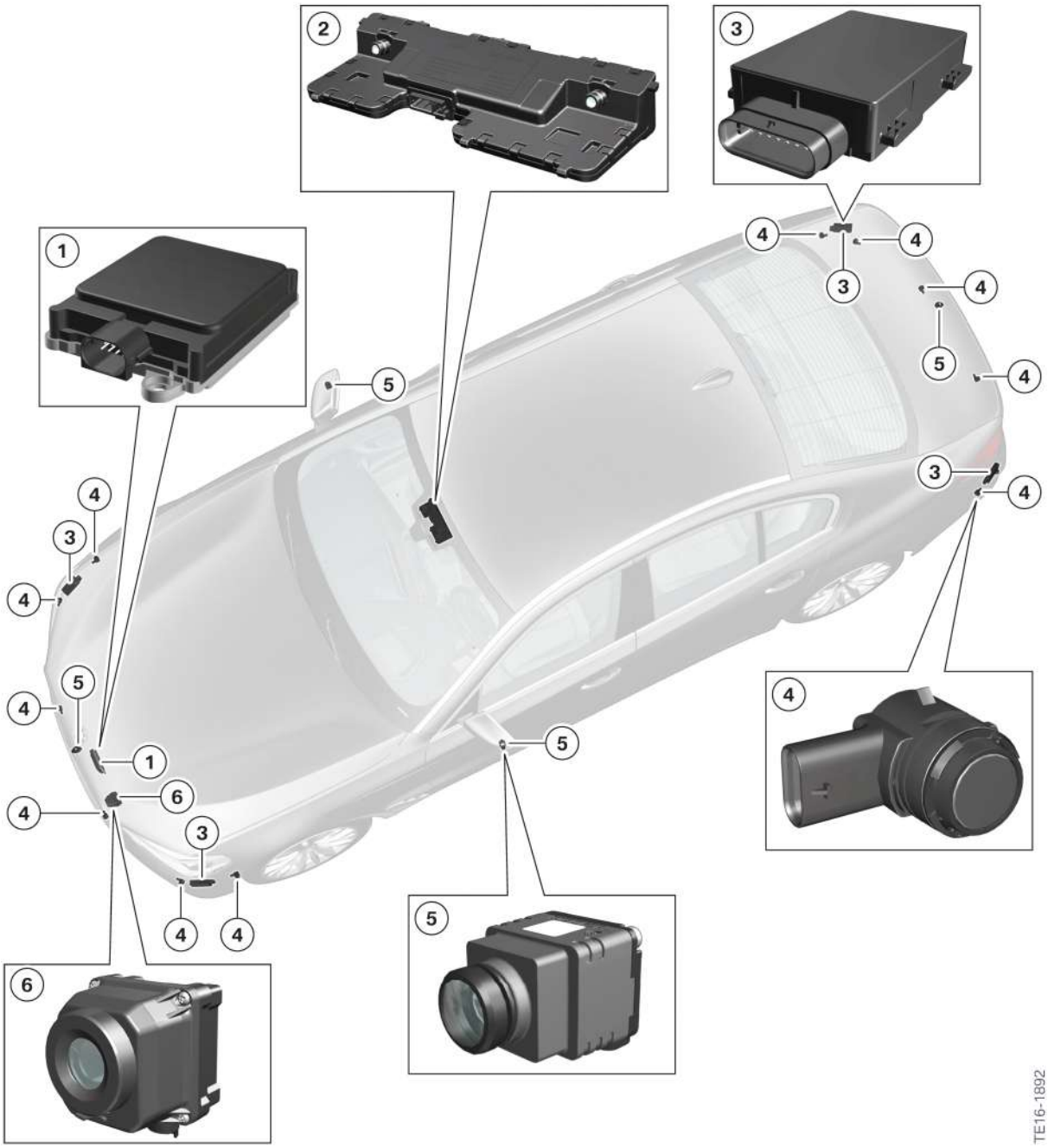
This product information explains the currently available assistance systems. The operation and functions of these systems will be described as from the F series vehicles. The functions of the assistance systems in older vehicles, such as the E series, are similar to those in the current vehicles.

The assistance systems are available as optional equipment in most vehicles. Depending on vehicle and year of manufacture, the assistance systems are available individually or in optional equipment packages.

1.4. Sensors

The following graphic shows an overview of the sensors currently used for the assistance systems at BMW. The graphic shows a current vehicle (G30). The corresponding table contains an overview where you can see which sensors are responsible for which assistance systems. Some assistance systems use several sensors simultaneously in order to guarantee more precise detection and measurement.

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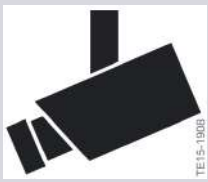


Overview of the sensors of the assistance systems

ASSISTANCE SYSTEMS.
1. INTRODUCTION.

Index	Assistance systems of the sensors
1	**Radar sensor, front centre:** • Active speed control with Stop&Go function • Traffic Jam Assist • Steering and lane control assistant • Tailgate warning • Junction warning • Avoidance assistant
2	**KAFAS camera:** • Lane departure warning • High-beam assistant • Tailgate warning • Active speed control with Stop&Go function • Traffic Jam Assist • Steering and lane control assistant • Junction warning • Avoidance assistant • Road sign recognition • Junction traffic sign recognition assistant • Wrong-way assistant • Lane change assistant
3	**Radar sensors, rear left and right:** • Lane change warning • Side collision warning • Crossing traffic warning • Avoidance assistant • Lane change assistant **Radar sensors, front left and right:** • Side collision warning • Crossing traffic warning • Avoidance assistant • Lane change assistant

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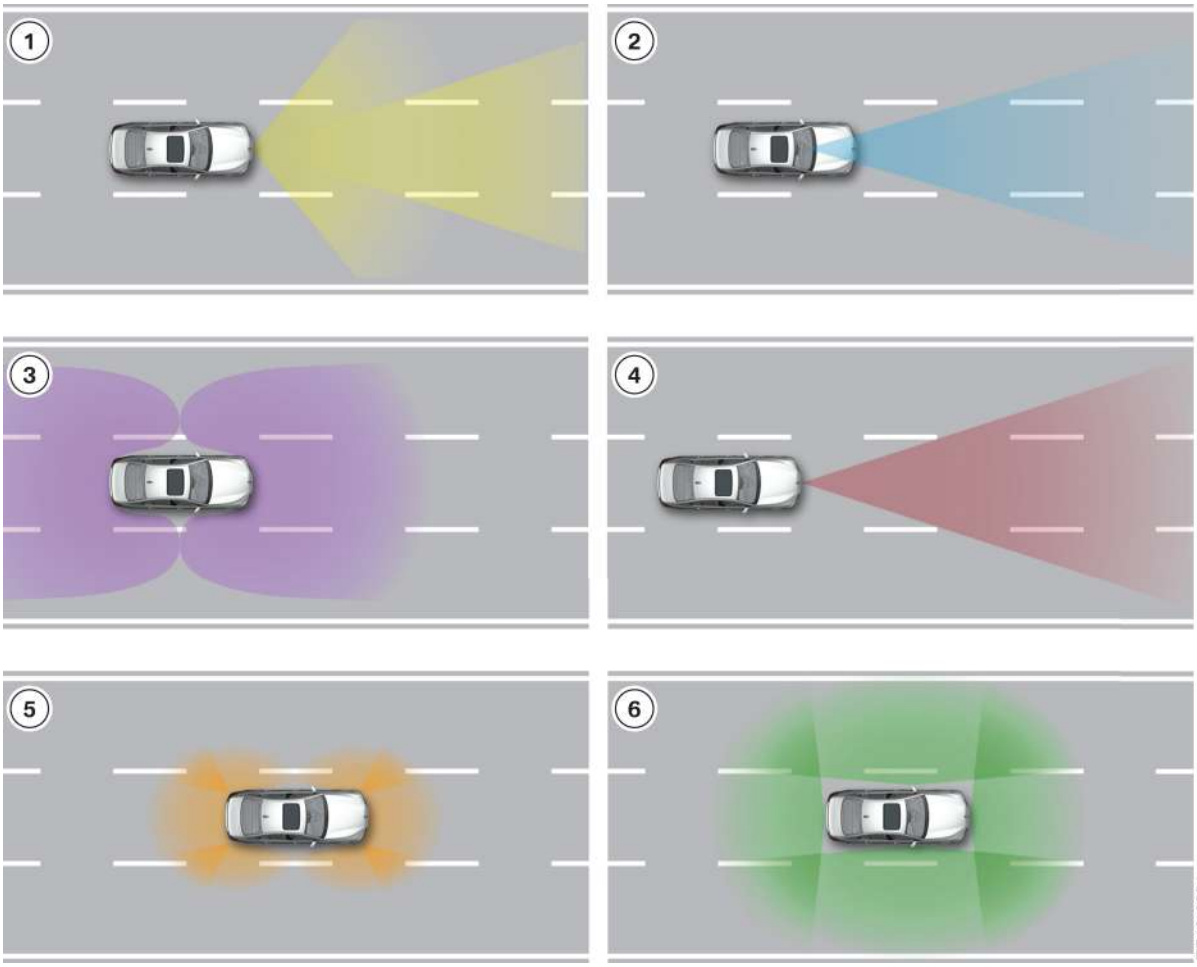
Index	Assistance systems of the sensors
4 	**Ultrasonic sensors** • Park Distance Control • Remote Control Parking • Parking manoeuvring assistant
5 	**Cameras** • Remote Control Parking • Reversing camera • Top View (270° view) • Surround View (360° view) • Remote 3D View
6 	**Night Vision camera:** • Night vision

1.4.1. Sensor detection range

The sensor detection ranges cover almost the complete area around the vehicle. The sensors used and the number of sensors depend on the equipment and vehicle.

The following graphic shows an overview of detection ranges of the individual sensors. The detection modes of the sensors are represented by different colours. The installation locations of the sensors and control units are vehicle-dependent. Further information on this can be found in the workshop information system.

ASSISTANCE SYSTEMS.
1. INTRODUCTION.



sensor detection range

Index	Explanation
1	ACC radar system
2	Camera system (KAFAS camera)
3	Radar systems
4	Night Vision camera
5	Ultrasonic systems
6	Camera systems

1.4.2. KAFAS camera

To facilitate understanding, only the KAFAS camera (KAFAS = camera-based driver assistance systems) is mentioned in this product information. 2 types of the KAFAS camera are used at BMW:

- KAFAS mono camera
- KAFAS stereo camera

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1. INTRODUCTION.



KAFAS camera comparison

Index	Explanation
1	KAFAS mono camera
2	KAFAS stereo camera (as from the G11)

A KAFAS stereo camera has been installed since the G11/G12. The KAFAS stereo camera offers the advantage that it not only enables the position of an object to be determined, but also its movement and direction of movement.

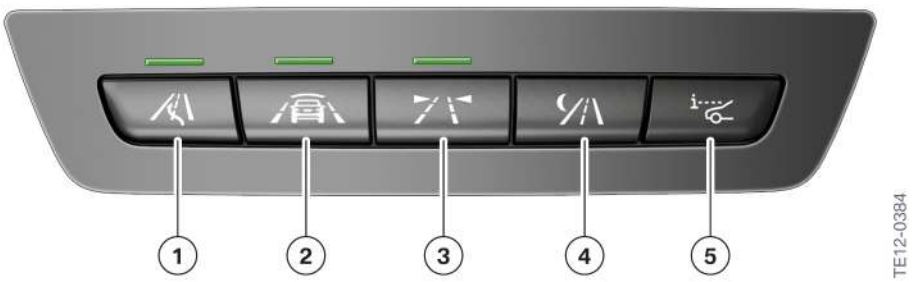
1.5. Operation

Operation of the individual assistance systems has be constantly further developed in the different vehicle generations.

In older vehicles, the individual assistance systems were activated by means of separate switches/ buttons. In the current vehicles, there is now an operating facility which can be used to operate most of the assistance systems, the intelligent safety button.

1.5.1. Operating facility for driver assistance systems

The F01 was equipped with a switch block which can be used to operate most of the assistance systems. The driver assistance system operating facility is located under the light operating unit.



Assistance systems operating unit

Index	Explanation
1	Lane change warning SWW
2	Intelligent Safety (collision warning with braking function and pedestrian warning)
3	Lane departure warning
4	Night vision
5	Head-Up Display HUD

1.5.2. Intelligent Safety button

The F15 was the first vehicle to be equipped with an intelligent safety button. Most of the assistance systems are operated by means of this button. The intelligent safety button is located in the instrument panel together with the hazard warning switch. The assistance systems can be personalised and also switched on and off by means of the button.



Intelligent Safety button

Operation

Operation is performed by pressing the intelligent safety button and in a menu for configuration of the assistance systems in central information display CID.

Press button:

- A screen is displayed on the central information display where the settings can be made. The individual settings are stored for the ID transmitter currently used.

Press button briefly:

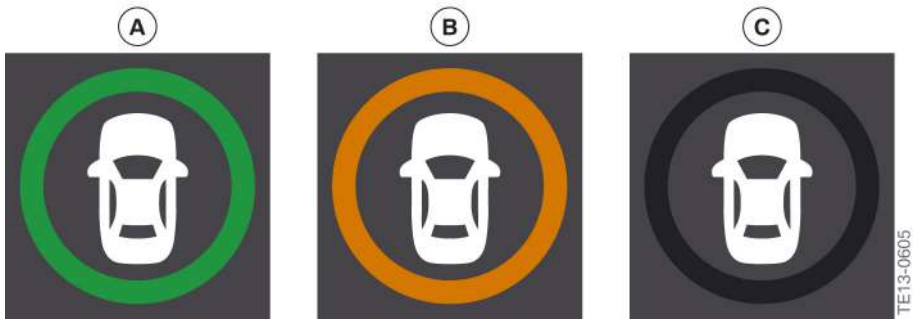
- The assistance systems are switched off individually, depending on the individual settings.
- The LED in the Intelligent Safety button lights up orange or goes out, depending on the Personal Profile.

Press button again:

- All assistance systems are switched on.
- The LED in the Intelligent Safety button lights up green.

Press button for an extended period:

- All assistance systems are switched off.
- The LED in the Intelligent Safety button goes out.



Intelligent safety button LED display

Index	Explanation
A	All assistance systems activated
B	Assistance systems with customised settings
C	All assistance systems deactivated

Assistance systems

Depending on the vehicle equipment, the following assistance systems are operated with the intelligent safety button:

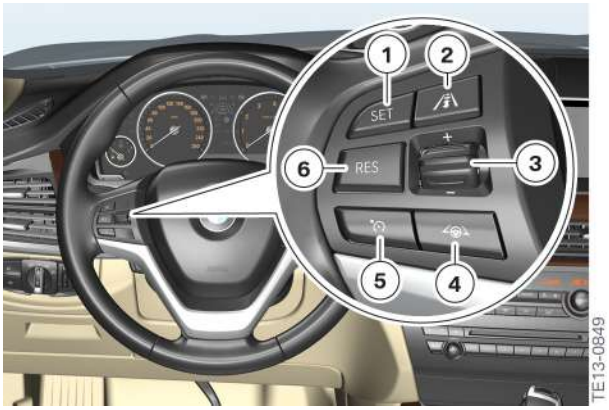
- Collision warning with city brake function
- Pedestrian warning with city braking activation function
- Night Vision with pedestrian and animal detection (personalisation only)
- Lane departure warning
- Lane change warning
- Side collision warning

1.5.3. Multifunction steering wheel

Some assistance systems are operated via the buttons on the multifunction steering wheel. These include:

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- Cruise control with braking function
- Active speed control with Stop&Go function
- Traffic Jam Assist
- Steering and lane control assistant
- Speed limit.



Multifunction steering wheel assistance systems

Index	Explanation
1	Set speed
2	Set distance
3	Set and change speed ±
4	Traffic Jam Assist
5	Switch cruise control on/off
6	Restore stored speed

1.6. System limits

All assistance systems have physical limits due to their sensor systems. The driver must therefore be able to intervene at any time, otherwise there is a risk of an accident.

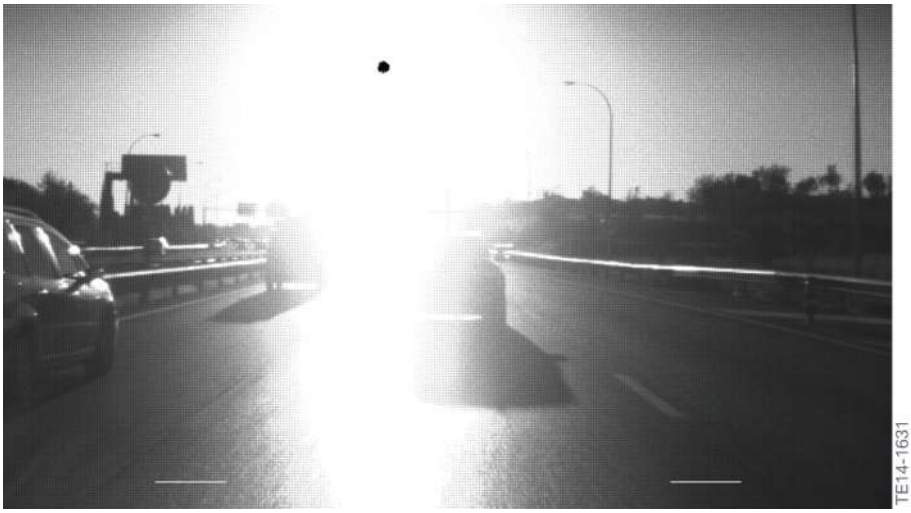
1.6.1. KAFAS camera

The function of the KAFAS camera can be restricted in the following situations, for example:

- Thick fog, heavy rain or snow
- Insufficient brightness
- strong back light
- Field of view of KAFAS camera dirty.

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Due to functional limitations and system restrictions it may transpire that warnings and orders are not issued, are issued too late or are unwarranted.



Example of functional limitation of KAFAS camera

1.6.2. Radar sensor

The function of the radar sensors can be restricted in the following situations, for example:

- Thick fog, heavy rain or snow
- on sharp bends
- Area of the radar sensors is dirty.

1.6.3. Ultrasonic sensors

The function of the ultrasonic sensors can be restricted in the following situations, for example:

- Thick fog, heavy rain or snow
- Sensors very dirty or iced up
- Area of the ultrasonic sensors is dirty
- if the objects are thin or wedge-shaped
- when the objects are low
- due to other ultrasound sources.

To ensure the ultrasonic sensors remain fully operational, they must be kept clean and free of ice. When using a high pressure cleaner, it must be ensured that the sensors are not damaged.

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2.1. Introduction

This section provides you with further information on the assistance systems which are used in the area of "Vision". These are:

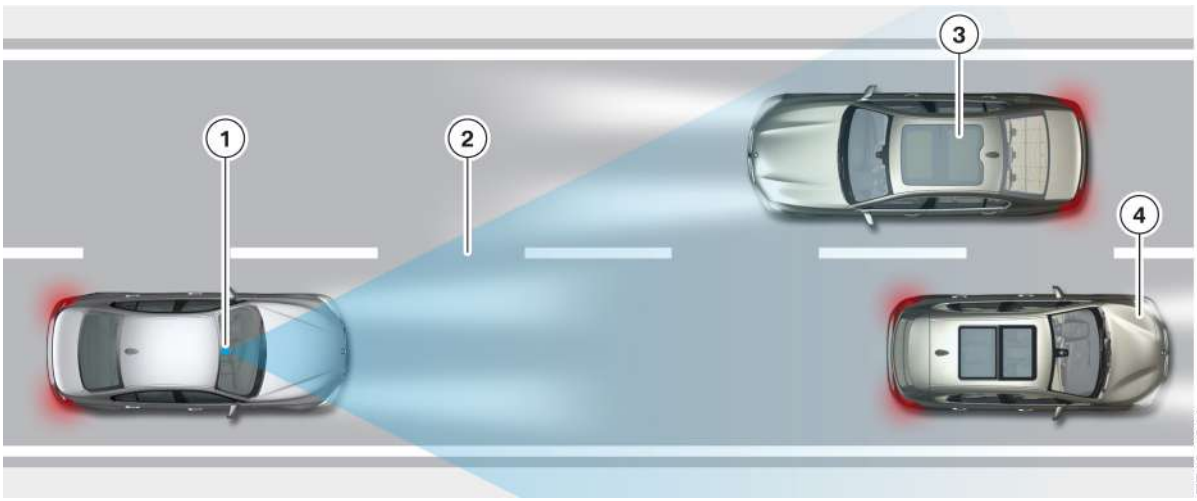
- High-beam assistant (FLA)
- Night Vision.

2.2. High-beam assistant

2.2.1. Function

The high-beam assistant (FLA) assists the driver in the use of high beam. The high-beam assistant switches high beam on and off depending on the situation and thus relieves the driver of the need to operate high beam. High beam can still be switched on and off manually by the driver as usual.

The system consists of a special camera, which is installed in the area of the windscreen, and a control unit. The camera records points of light when driving which are evaluated in the control unit. The camera can distinguish between different parameters such as light colour and light intensity. The high beam is activated automatically if all prerequisites are met. The high beam is automatically dipped only when a switch-off recommendation is detected. The high beam is automatically dipped if there are oncoming vehicles or vehicles driving ahead.



High-beam assistant function

Index	Explanation
1	KAFAS camera
2	Detection range of KAFAS camera
3	Oncoming vehicle
4	Vehicle ahead

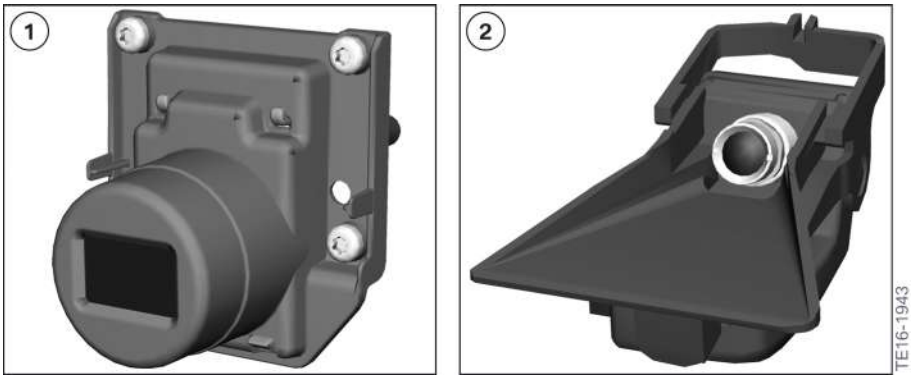
ASSISTANCE SYSTEMS.
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High-beam assistant indicator light

2.2.2. Sensors

Depending on the vehicle equipment, 2 different cameras are used for the function. If the vehicle has an assistance system for which a KAFAS camera is already needed, the high-beam assistant function is realised using this camera. If the vehicle only has a high-beam assistant, a separate camera is installed in the interior mirror.



High-beam assistant camera

Index	Explanation
1	High-beam assistant camera in the interior mirror
2	KAFAS camera

2.2.3. Operation

The high-beam assistant is operated by means of the turn signal/high beam switch. The light switch must be in the position “A” (automatic driving lights control).



Activation of high-beam assistant

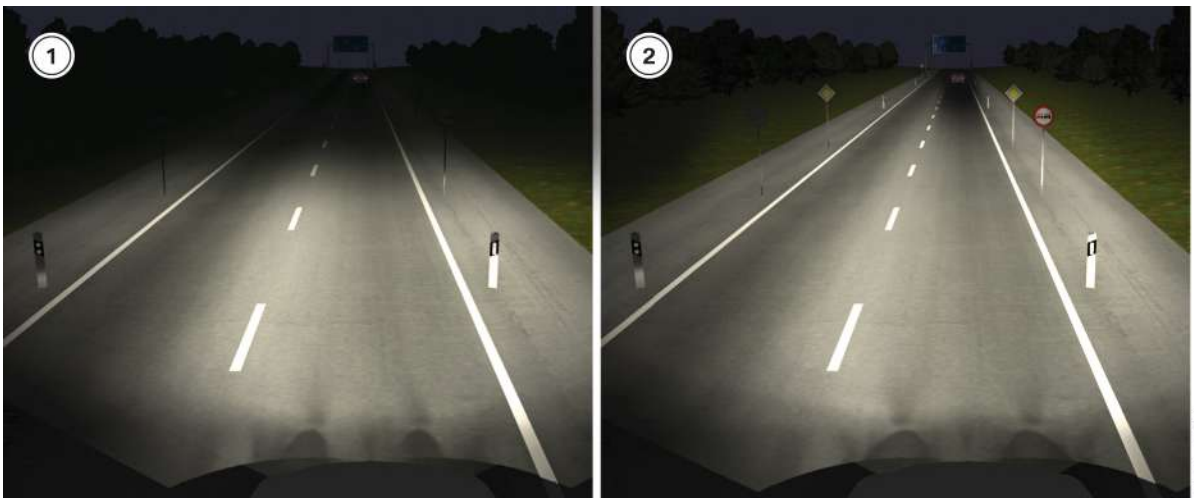
Index	Explanation
1	Light switch
2	High-beam assistant button on the turn signal/high beam switch

2.2.4. Dazzle-free high-beam assistant

A dazzle-free version of the high-beam assistant has been available since the F30. In combination with adaptive headlights, the light cone is adjusted so that oncoming vehicles and vehicles driving ahead are not dazzled.

The high beam is not completely switched off, but only the areas are suppressed which would dazzle the oncoming vehicle or vehicle in front. The high beam is not completely switched off in this case and the blue high beam indicator light remains on.

Operation of the dazzle-free high-beam assistant is identical to that for the conventional high-beam assistant.

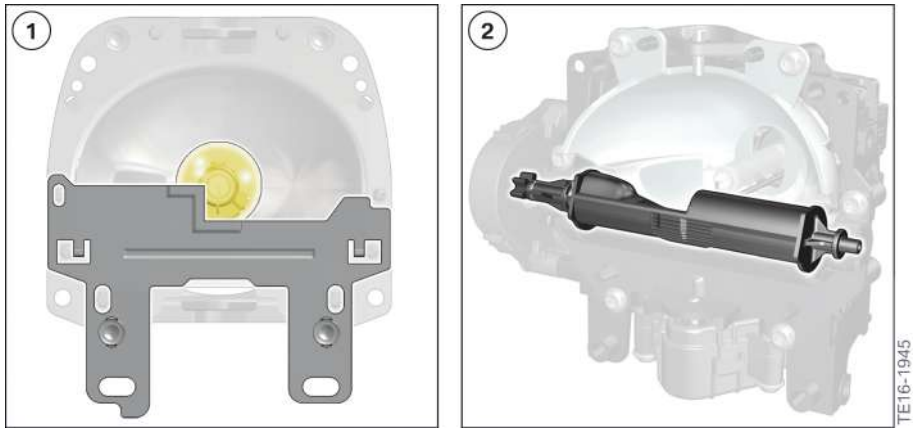


Anti-glare high-beam assistant

Index	Explanation
1	High-beam headlight dipped
2	High beam active, vehicle driving ahead omitted from light cone

Other road users are omitted from the main beam cone by special masking elements on the headlights and horizontal swivelling of the headlights. Vehicle driving in front and oncoming vehicles are omitted from the light cone.

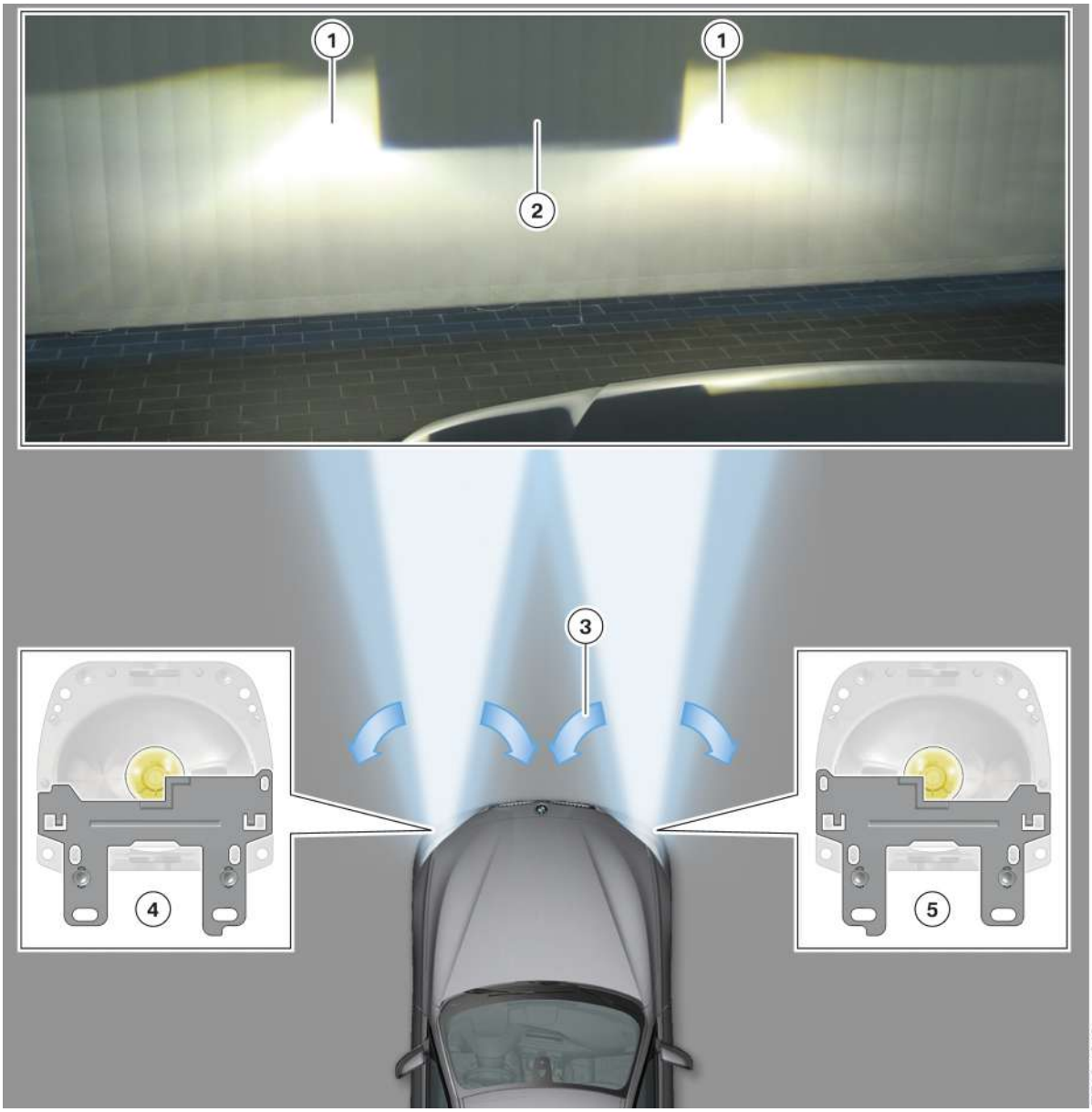
ASSISTANCE SYSTEMS.
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Headlight of dazzle-free high-beam assistant

Index	Explanation
1	Masking element on xenon headlight F30
2	Masking element on xenon headlight F01/F02/F07/F10 LCI (cylinder)

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High beam of dazzle-free high-beam assistant

Index	Explanation
1	High beam cone
2	Area of high beam which is not illuminated due to masking element
3	Headlight swivelling
4	Masking element on left headlight
5	Masking element on right headlight

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2.3. Night vision

2.3.1. Function

Night Vision provides the driver with a black-and-white image of the vehicle surroundings in front of the vehicle on the central information display CID by means of a thermal imaging camera. Night Vision is a passive system without active infrared illumination.

The higher the temperature of the objects in front of the vehicle, the brighter they appear on the display. This allows the driver to recognise heat-radiating objects such as pedestrians or animals early on.

2.3.2. Sensors

The Night Vision system consists of the Night Vision camera and a control unit. The Night Vision camera is installed in the radiator grille.



Night Vision camera

The picture signals are transmitted to the Night Vision electronics via an LVDS data line.

2.3.3. Operation

Night Vision is switched on and off by means of a button on the light operating unit.

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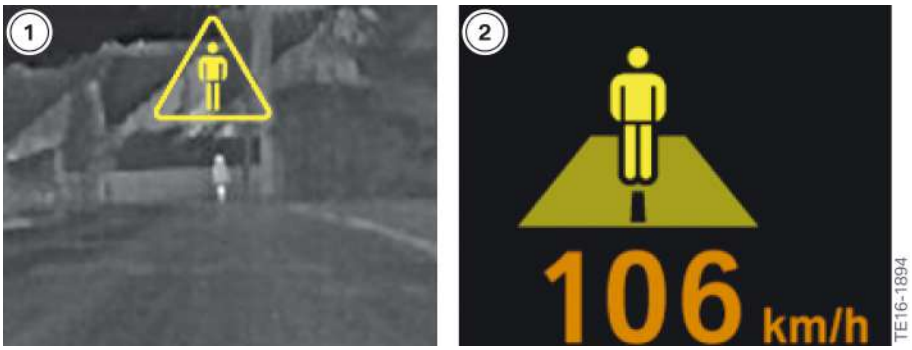
Night Vision button

Index	Explanation
1	Night Vision button

2.3.4. Warnings

Initially, Night Vision was able to display only image content on the CID. The Night Vision system has been extended to include new functions with very further development.

With the F01, the Night Vision system additionally included pedestrian detection. The system warns the driver about persons located in a certain area in front of the vehicle. So that the driver is not distracted unnecessarily, a warning is issued only if the person is within a certain area or moves in the direction of this area.



Pedestrian warning display

Index	Explanation
1	Warning on the central information display CID
2	Warning on the Head-Up Display HUD

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With the F01 LCI, the pedestrian warning function was extended by a dynamic marker light. Here, a pedestrian located in the corresponding area is illuminated by a high-power marker light.



Vehicle without and without dynamic marker light

Depending on vehicle, the dynamic marker light is integrated in different light sources:

Vehicle	Installation location of dynamic marker light
F0x F1x	Instead of fog lights
G11 G30, G31	Integrated in the headlight

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2.3.5. Zusammenfassung

Vehicle	Function	Special features
Since the F01	Pedestrian warning 	Provides a warning about pedestrians located in the area directly in front of the vehicle. The warning is issued only on the central information display CID.
Since the F01 LCI	Acute pedestrian warning 	Provides a warning about pedestrians located in the warning area in 2 stages. The warning is issued either on the instrument cluster or, where applicable, in the Head-Up Display HUD. Detected persons are illuminated by a high-power marker light.
Since the F10 LCI	Acute pedestrian warning  Animal warning 	Provides a warning about pedestrians located in the warning area in 2 stages. The warning is issued either on the instrument cluster or, where applicable, in the Head-Up Display HUD. Detected persons are illuminated by a high-power marker light. Warns about animals located in the warning area.
Since the G11/G12	Acute pedestrian warning  Acute animal warning 	Provides a warning about pedestrians located in the warning area in 2 stages. The warning is issued either on the instrument cluster or, where applicable, in the Head-Up Display HUD. Detected persons are illuminated by a high-power marker light. Provides a warning about animals located in the warning area in 2 stages.

BMW Night Vision is offered only for vehicles as from the BMW 5 Series.

2.3.6. Service

The following functions can be adjusted by the service organisation:

Adjustment of Night Vision camera

The Night Vision camera can be mechanically adjusted in axial direction.

Adjustment of headlights for dynamic marker light

The correct setting of the headlight for the dynamic marker light is imperative for fault-free functioning of the dynamic marker light. The headlight adjusting device “MAHA Lite 3” should be used for this.

2.4. MINI

The MINI models are equipped with the following assistance systems from the "Vision" area:

- High-beam assistant.

ASSISTANCE SYSTEMS.
3. PARKING.

3.1. Introduction

This section provides you with further information on the assistance systems which are used in the area of "Parking". These are:

- Park Distance Control (PDC)
- Parking manoeuvring assistant PMA
- Remote Control Parking
- Camera systems.

3.2. Park Distance Control

3.2.1. Function

Park Distance Control PDC warns about obstacles located in front of or behind the vehicle. This makes it easy to manoeuvre or park when space is tight.

PDC has been offered by BMW since 1992. 2 different versions area offered in current vehicles:

- PDC rear (OE 507)
- PDC front and rear (OE 508).

If a CID is installed in the vehicle, the detected obstacles will be indicated both visually and acoustically. In the case of vehicles without CID, only an acoustic warning will be provided.

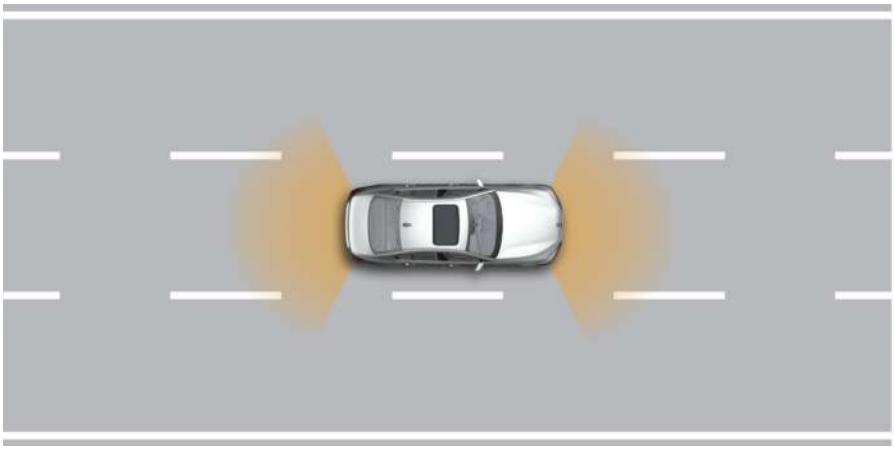


Park Distance Control display on central information display

3.2.2. Sensors

Ultrasonic sensors are used for the sensor system. Depending on the equipment, 4 ultrasonic sensors are installed in the rear bumper and 4 ultrasonic sensors in the front bumper. An exception here is the F01/F02, which has 5 ultrasonic sensors installed in the front bumper.

ASSISTANCE SYSTEMS.
3. PARKING.



Park Distance Control sensors

3.2.3. Operation

PDC can be activated manually by means of the PDC button in the centre console. PDC is also activated automatically when reverse gear is engaged.



Park Distance Control button

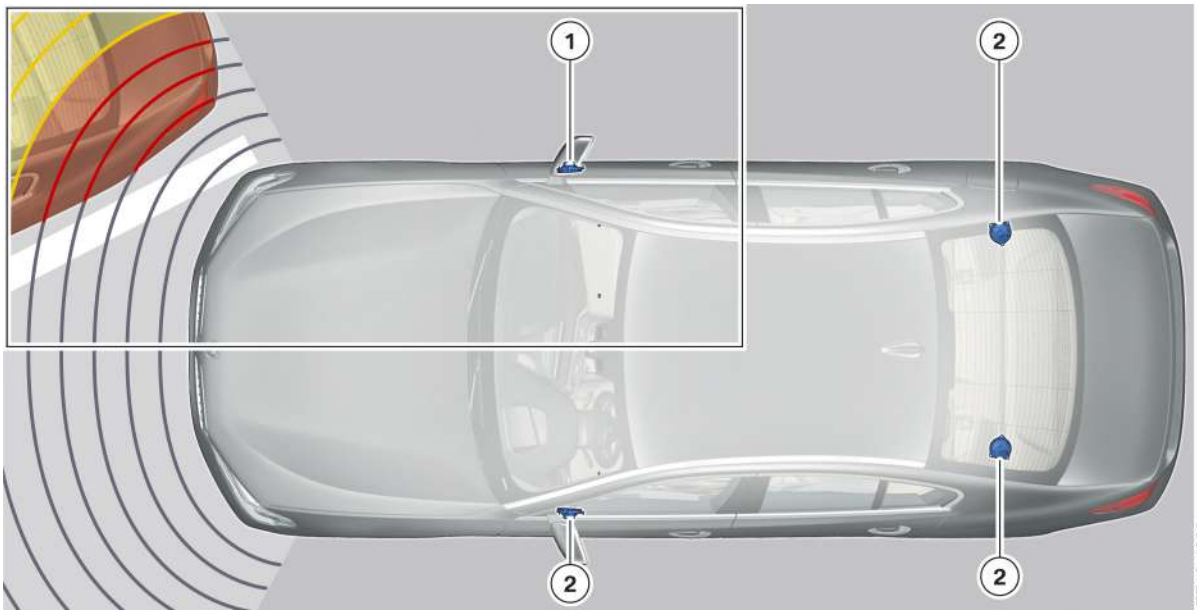
Index	Explanation
1	Park Distance Control (PDC) button

ASSISTANCE SYSTEMS.
3. PARKING.

3.2.4. Warning

A warning is provided by acoustic signal tones. These are output by the corresponding speakers to permit better recognition. If an obstacle is located at the front right, for example, the acoustic signal will be output via the front right speaker.

In vehicles which are equipped with a central information display CID, a visual warning is provided in addition to the acoustic signal.



Acoustic warning

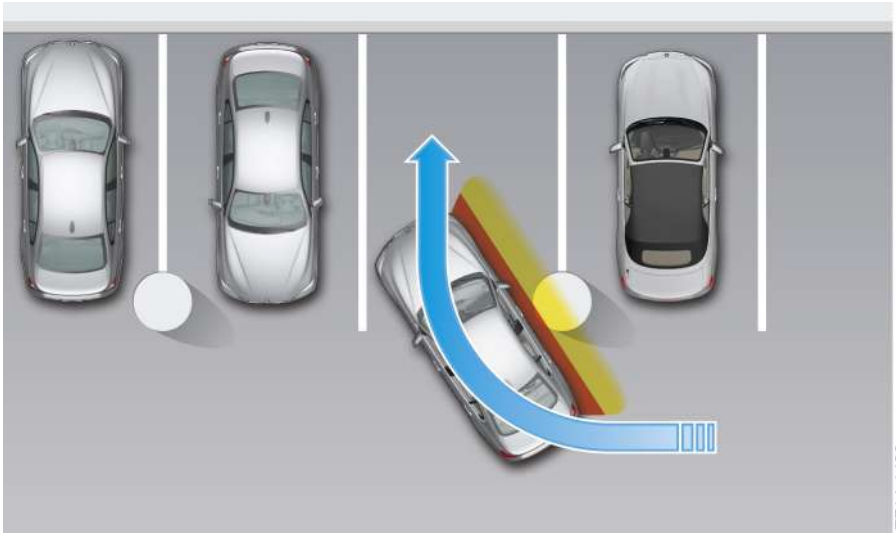
Index	Explanation
1	Speaker with warning signal
2	Speaker without warning signal

3.2.5. Side protection

With the side protection, the side area of the vehicle is now also monitored in addition to the front and rear. The side protection has been installed since March 2016. Side protection is **not** separate optional equipment, but is included in vehicles which are equipped with a Parking Manoeuvring Assistant PMA and a Professional navigation system (OE 609).

The side ultrasonic sensors of the PMA are used to provide the side protection function.

ASSISTANCE SYSTEMS.
3. PARKING.



Principle of side protection

The side protection is active at the same time as PDC and functions up to a speed of 10 km/h.

The measuring range around the B-pillar and parts of the doors is not monitored by the ultrasonic sensors. Instead this area is imaged using a computer model while the vehicle drives past.

The side protection is not active during a parking operation with the PMA.

An acoustic warning is only issued if the detected obstacle is lying in the path of the vehicle. A continuous alarm is output once the obstacle is closer than 25 cm away.



Side protection indication in the central information display (CID)

3.2.6. Auto PDC

Auto PDC was used for the first time in the F56 at MINI or in the F15 at BMW. PDC is automatically activated under the following conditions when driving forwards:

- if there is an object closer than 60 cm to the front of the vehicle when driving forwards
- if there is an object closer than 1.5 m to the rear of the vehicle when reversing
- Speed below 5 kph.

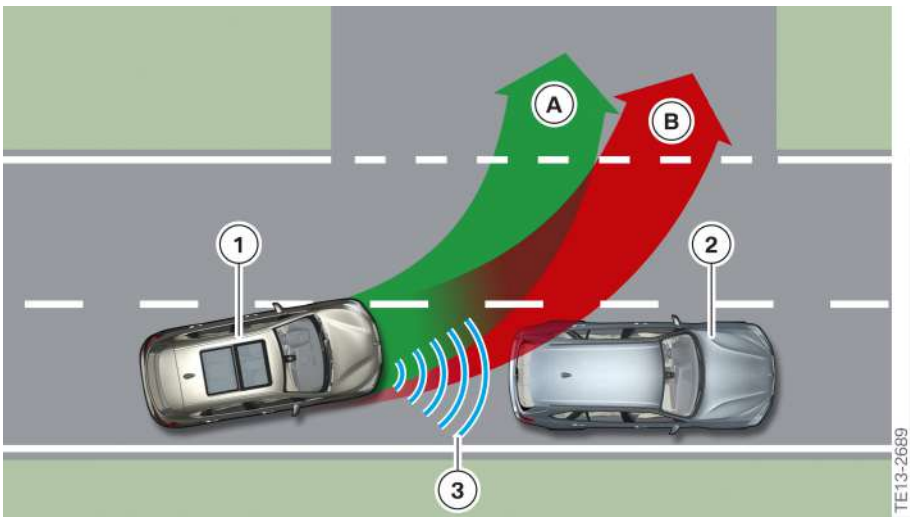
ASSISTANCE SYSTEMS.
3. PARKING.

The automatic switching-on upon the identification of obstacles can be switched on and off in the "Settings" menu using a controller.



Auto PDC setting in the central information display CID

A sound is output only if the detected object (in front of or behind the vehicle) is directly in the lane (risk of collision). If the detected object is not directly in the lane, the acknowledgement is only effected visually in the CID.



Auto PDC object detection

Index	Explanation
A	Lane without object contact (visual indication in the CID)
B	Lane with object contact (visual and acoustic indication)
1	Object turning off
2	Detected object
3	Detection of obstacle by the PDC sensors

If an object was detected by Auto PDC, the system must be reactivated so that it cannot be switched on accidentally, e.g. in a car wash. Renewed activation is performed by the following parameters:

- Driving speed > 5 km/h
- drive position change

There is no automatic activation of the PDC in the “neutral” drive position in order to avoid switching on here also, e.g. in car washes.

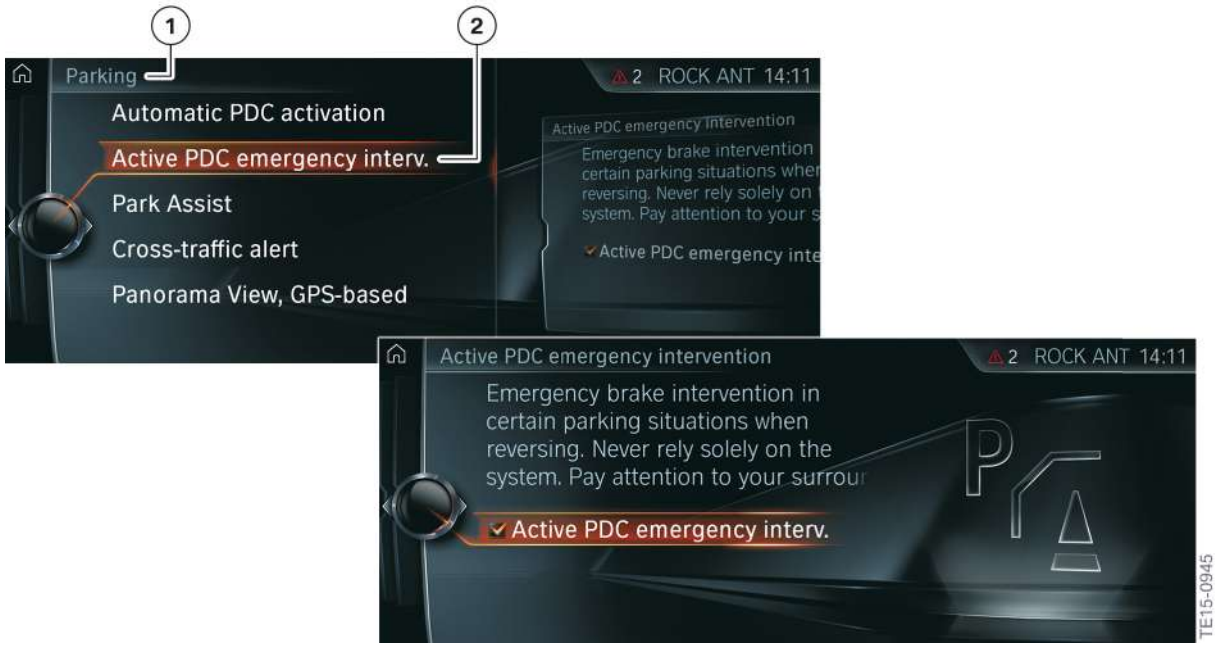
3.2.7. Active Park Distance Control

Active Park Distance Control was used for the first time in the G11/G12. This automatically brakes the vehicle as soon as an obstacle is detected behind the vehicle. The speed of the vehicle must not be more than 6 km/h.

If the accelerator pedal is pressed and the obstacle is close, the brakes are preconditioned and gentle initial braking is initiated as a warning for the driver. If the accelerator pedal is not pressed and the vehicle is very close to the obstacle, the function brakes the vehicle with the maximum possible brake force until it comes to a standstill.

The automatic brake intervention can be overridden at any time by pressing the accelerator pedal.

Active PDC can be switched on and off in the “Settings” menu by a corresponding controller.



Active PDC on the central information display CID

Index	Explanation
1	"Parking" menu
2	Active PDC emergency intervention

3.2.8. Control unit

The PDC function is accommodated in various control units in the vehicle. Refer to the following table for the corresponding information:

Vehicle	Control unit for PDC function
From F01	Function integrated in the PDC control unit.
From F10	Function integrated in the PDC control unit. In vehicles with the optional equipment PMA, the function is integrated in the PMA control unit.
From F20	Function integrated in the Rear Electronic Module REM control unit. In vehicles with PMA, the function is integrated in the PMA control unit.
From F15	Function integrated in the Body Domain Controller BDC control unit. In vehicles with PMA, the function is integrated in the PMA control unit.

3.3. Parking manoeuvring assistant

3.3.1. Function

The Parking Manoeuvring Assistant PMA relieves the strain on the driver in two ways. On the one hand, the PMA decides independently whether a parking space is large enough. On the other, PMA assist the driver by means of steering interventions during the parking procedure.

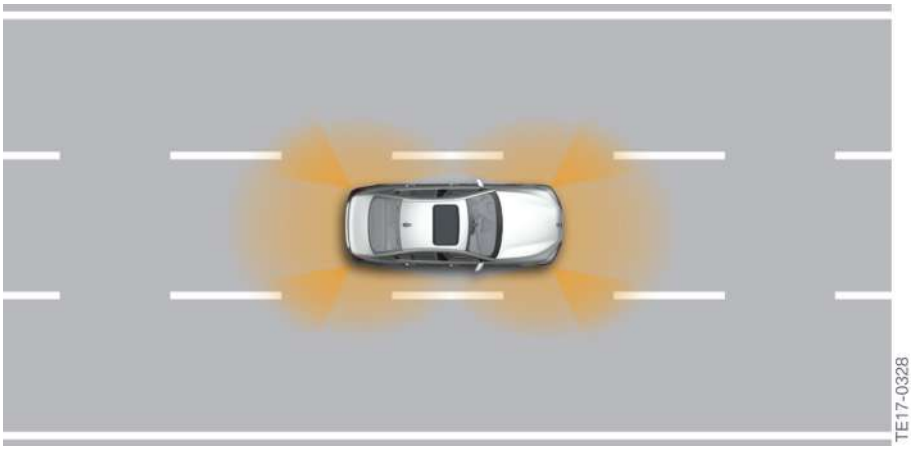
The Parking Manoeuvring Assistant PMA was offered for the first time in the F10. The first generation of the PMA was able to provide assistance only for parking spaces located parallel to the road. It is now also possible to provide assistance for parking spaces that are perpendicular to the road.

The PMA system measures potential parking spaces when driving by at speeds under approx. 35 km/h, regardless of whether PMA has already been activated or not. When PMA is activated, the driver is provided with a visual indication in the CID if a suitable parking space is detected. The Parking Manoeuvring Assistant then performs steering for the actual parking procedure. The driver remains responsible for acceleration and deceleration of the vehicle. As from the G11/G12 , the vehicle does not just assist with lateral guidance but also with longitudinal guidance.

3.3.2. Sensors

Additional ultrasonic sensors are required for detection of parking spaces for PMA. The additional ultrasonic sensors are installed at the sides of the front bumper.

ASSISTANCE SYSTEMS.
3. PARKING.



Ultrasonic sensors of Parking Manoeuvring Assistant

3.3.3. Operation

There are 2 options for activating the PMA system:

- Activation via the parking assistance button in the centre console
- Activation by engaging reverse gear and then operation via controller.



Parking Manoeuvring Assistant button in centre console

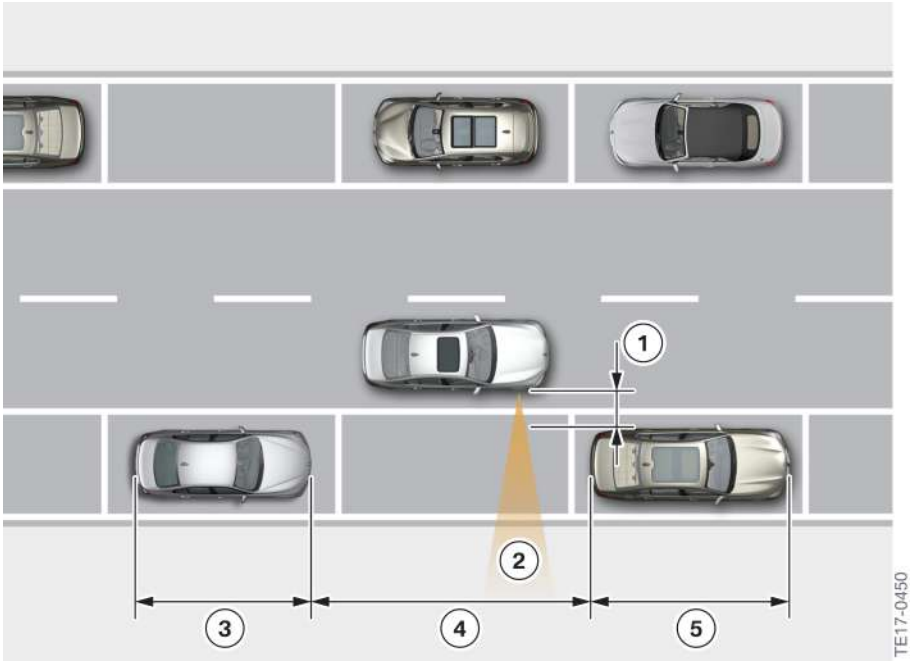
Index	Explanation
1	Button for Parking Manoeuvring Assistant PMA

ASSISTANCE SYSTEMS.
3. PARKING.

3.3.4. Kerbside parking

1st generation

A parking space is detected only if it is between 2 objects.



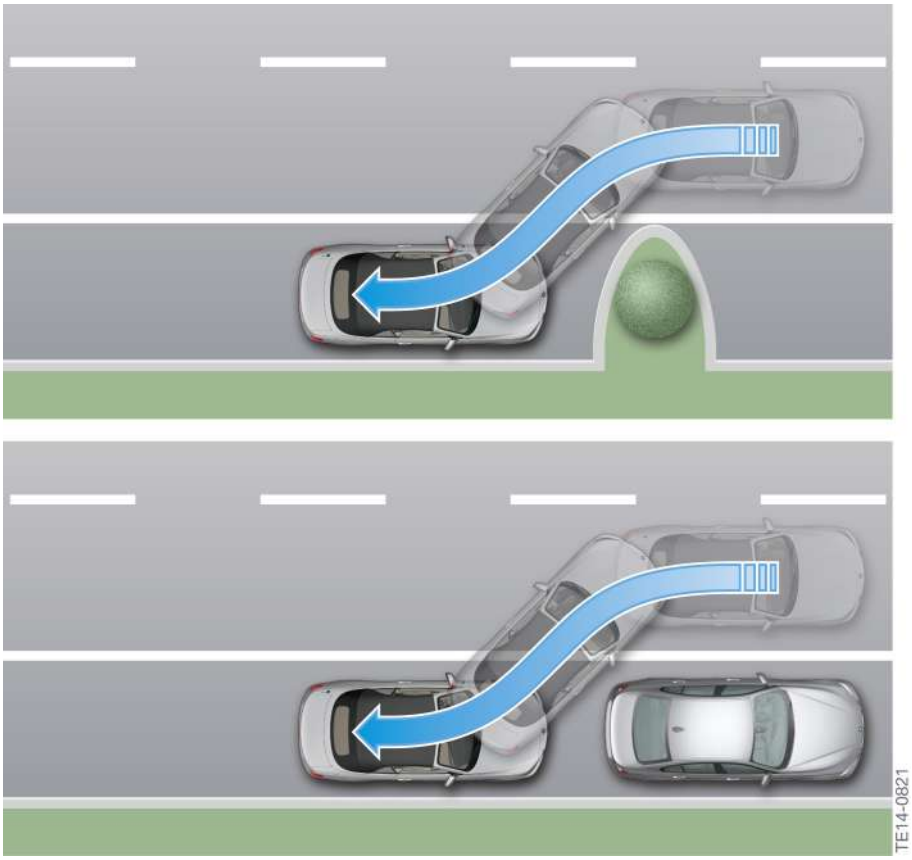
Requirements on the parking space

Index	Explanation
1	Maximum distance to the row of parked vehicles: 1.5 m
2	Horizontal opening angle of the ultrasonic sensor: ± 10°, range approx. 4.5 m
3	Vehicle or object length: min. 1.5 m
4	Length of the parking space; vehicle length: + approx. 1.2 m
5	Vehicle or object length: min. 1.5 m

2nd generation

The 2nd generation (from F23) of the Parking Manoeuvring Assistant PMA no longer needs 2 objects for detection of a parking space. Instead, one object is sufficient.

ASSISTANCE SYSTEMS.
3. PARKING.



Principle of longitudinal parking with 2nd generation system

The following table shows the changes in parking space search between first and second generation PMA:

Requirements	Parking Manoeuvring Assistant before 03/15	Parking Manoeuvring Assistant after 03/15
Size of parking space	Vehicle length + approx. 1.2 m	Vehicle length + approx. 0.8 m
Length of objects	min. 1.5 m	min. 0.5 m
Minimum number of objects	2	1
Parking space	Parallel	Parallel and perpendicular

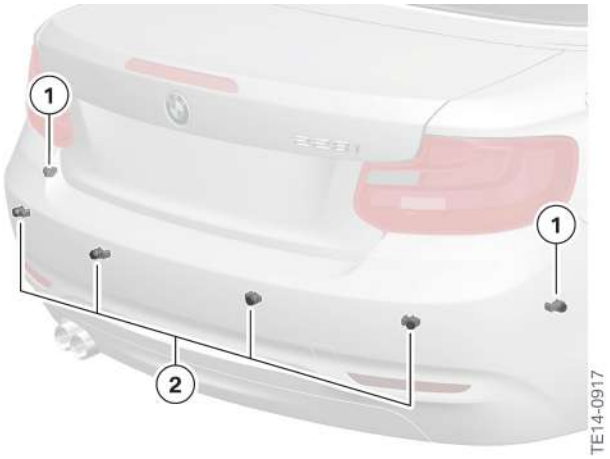
3.3.5. Cross & in-line parking spots

The 2nd generation PMA can also park in parking spaces which are perpendicular to the road.

Sensors

Two additional ultrasonic sensors are installed in the rear bumper to precisely detect the size of a perpendicular parking space during the parking manoeuvre. These ultrasonic sensors are needed to determine the distance from nearby objects during a parking procedure.

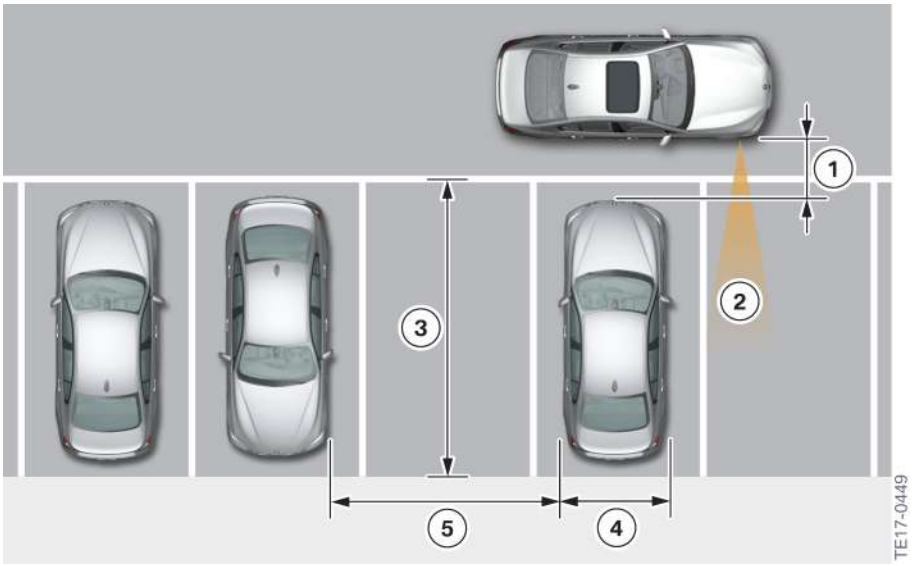
ASSISTANCE SYSTEMS.
3. PARKING.



Additional ultrasonic sensors in the rear bumper

Index	Explanation
1	Additional ultrasonic sensors at the sides of the rear bumper
2	ultrasonic sensors in the rear bumper

Requirements on the parking space



requirement, parking space, cross parking

Index	Explanation
1	Maximum distance to the row of parked vehicles: 1.5 m
2	Opening angle of the ultrasonic sensor: horizontal ± 40°, range approx. 4.2 m
3	Minimum depth of parking space: own vehicle length
4	Vehicle or object width: min. 0.5 m
5	Width of the parking space: vehicle width + 0.7 m up to max. 5 m

ASSISTANCE SYSTEMS.
3. PARKING.

The depth of perpendicular (cross) parking spaces must be gauged by the driver himself. Due to technical limits the system can only approximately determine the depth of perpendicular (cross) parking spaces.

3.3.6. Longitudinal and lateral guidance

As from the G11/G12, the PMA is also able to control the longitudinal guidance of the vehicle in addition to the lateral guidance. This function is available only in vehicles with an automatic transmission. In order to ensure the presence of the driver during the parking manoeuvre, it is necessary to press and hold the parking assistance button for the entire parking procedure.



Parking assistance button

Index	Explanation
1	Parking assistance button

3.3.7. Service

Cancellation of parking procedure

The parking procedure is cancelled in the following situations:

- Max. parking speed of 10 km/h has been exceeded
- Wrong turn indicator selection (e.g. left instead of right)
- Intervention by the driver

- A door is opened
- Selection of wrong gear
- DSC identifies slip.

Exchange of control unit or sensor

After the PMA control unit or an ultrasonic sensor is replaced, no start-up is required. The control unit checks at every new start whether the software of the ultrasonic sensors is compatible with the control unit software. If this is not the case, the following service function must be performed using the workshop information system: “Software update for ultrasonic sensors”.

The PMA control unit monitors itself for faults and enters corresponding fault code entries if required. There may be customer complaints if the ultrasonic sensors are not installed correctly, e.g. after an accident, but no faults will be stored:

- Parking spaces are not longer detected
- During a parking manoeuvre, the vehicle drives very close to or a long way past the front vehicle
- After parking, the vehicle is very close to or far away from the kerb
- The vehicle is parked in the gap at an angle.

3.4. Remote Control Parking

3.4.1. Function

The system permits remote-controlled manoeuvring of the vehicle into and out of a parking space using the BMW display key. The system therefore assists the driver in parking in tight parking bays (such as in garages or multi-storey car parks), where it is difficult for the driver to get in and out of the vehicle.

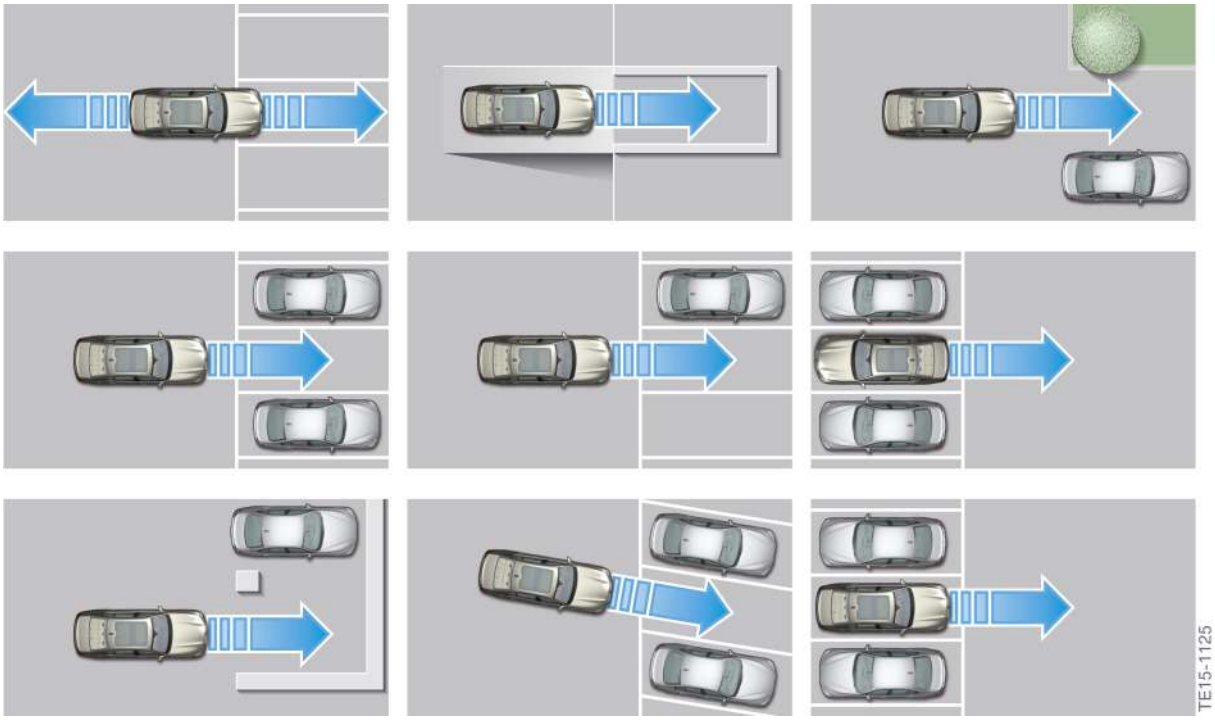
The parking assistant was used for the first time in the G11.

The system performs the following vehicle functions (taking into account the system limits):

- Longitudinal guidance
- Lateral guidance (minimum steering correction)
- Starting and switching off the combustion engine
- Assistance in monitoring the vehicle surroundings
- Switching the low-beam headlights on and off
- Releasing and engaging the electric parking brake

The graphic below shows an overview of the possible situations in which the remote-controlled parking function can be used:

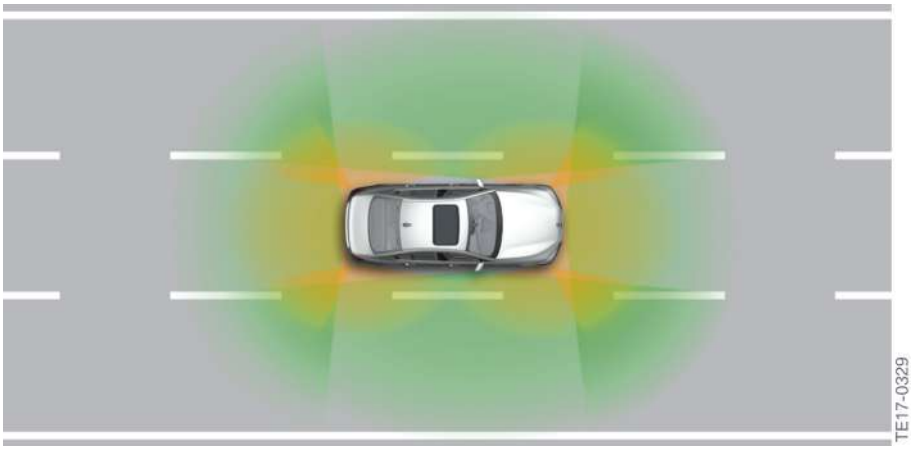
ASSISTANCE SYSTEMS.
3. PARKING.



Overview of possible Remote Parking use cases

3.4.2. Sensors

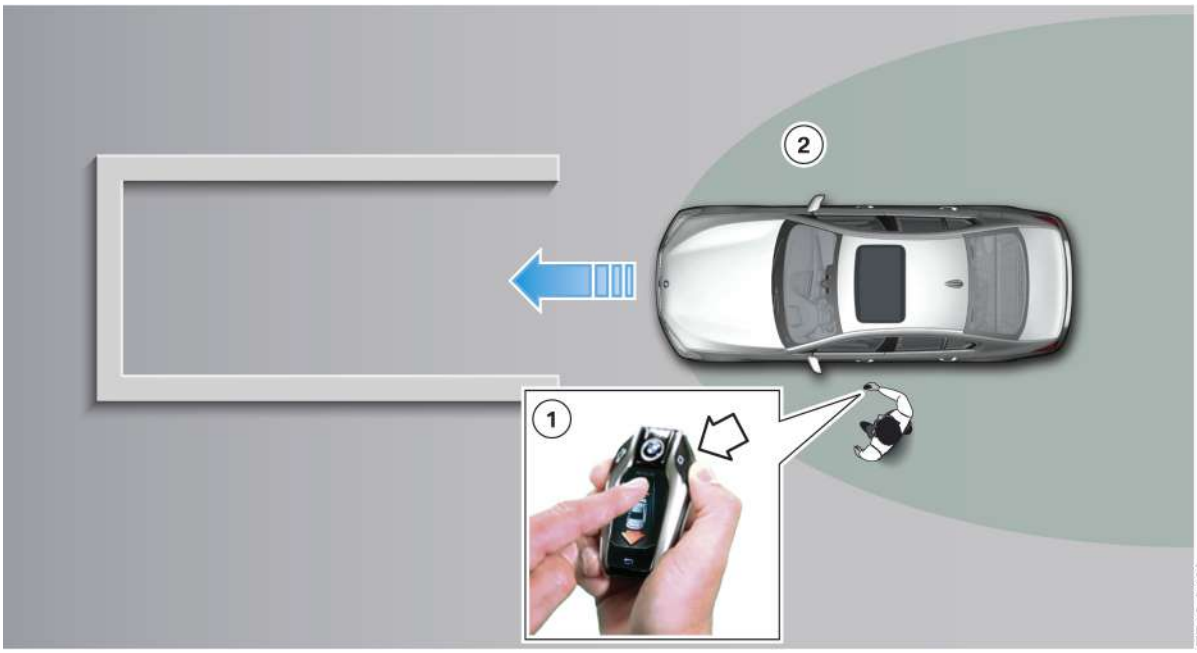
The entire parking procedure is monitored by the ultrasonic sensors and the Surround View cameras.



3.4.3. Operation

The parking assistant is operated by means of the BMW display key. The remote-controlled parking button must be pressed and held throughout the entire parking procedure. The parking procedure itself is started on the touch display of the BMW display key.

ASSISTANCE SYSTEMS.
3. PARKING.



Principle of parking with the parking assistant

Index	Explanation
1	BMW display key
2	Operating range for parking assistant

3.5. Camera systems

This section provides further information on the camera systems used at BMW:

- Reversing camera
- Side View
- Top View
- Panorama View
- Surround view
- Remote 3D View.

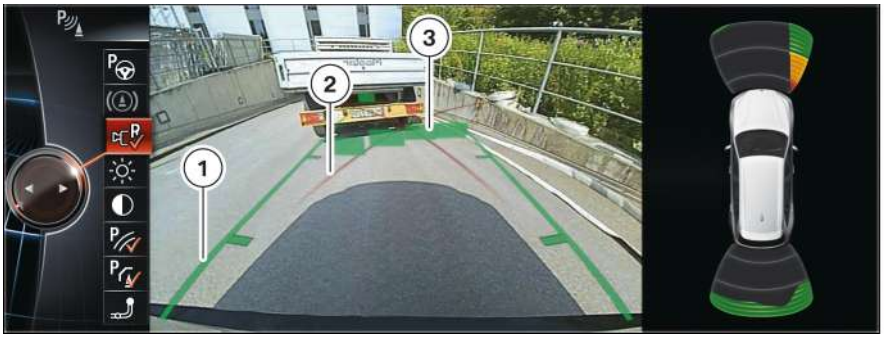
3.5.1. Rear view camera

Function

The main function of the rear view camera system is optical detection of a wide-angled view of the rear area of the vehicle in order to offer the driver additional support during the parking procedure or manoeuvring. The image is forwarded from the rear view camera via a CVBS line to the headunit.

The image from the rear view camera is displayed with additional auxiliary lines in the CID.

ASSISTANCE SYSTEMS.
3. PARKING.



Rear view camera display on the central information display

Index	Explanation
1	Lane line
2	Turning circle line
3	Obstacle marking

The turning circle line shows the path of the smallest possible turning radius on a level roadway.

The driving lane line helps the driver assess the space requirements when parking and manoeuvring on a level roadway. The lane line depends on the current steering angle and is adapted continuously to the steering wheel movements.

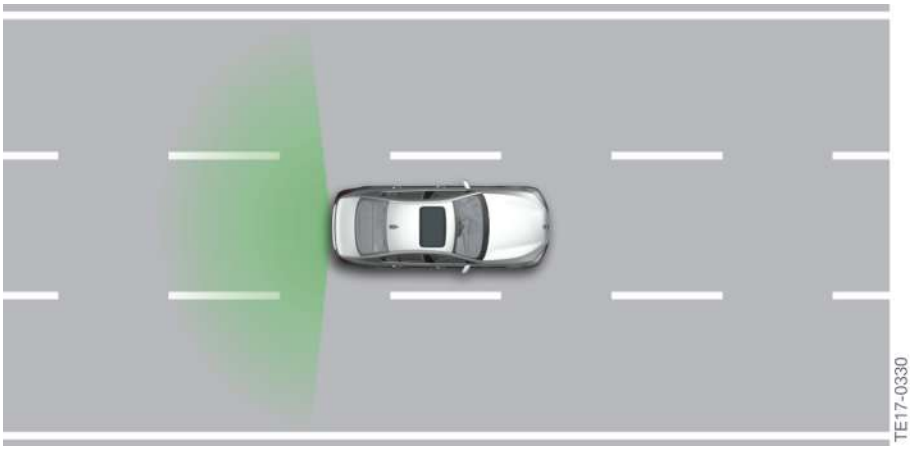
The view of the rear view camera can be changed via the controller. For example, a towbar zoom is also available.



Towbar zoom

ASSISTANCE SYSTEMS.
3. PARKING.

Sensors



Rear view camera sensor

Operation

The rear view camera is activated either by engaging reverse gear or pressing the PDC button.



Park Distance Control button

Index	Explanation
1	Park Distance Control button

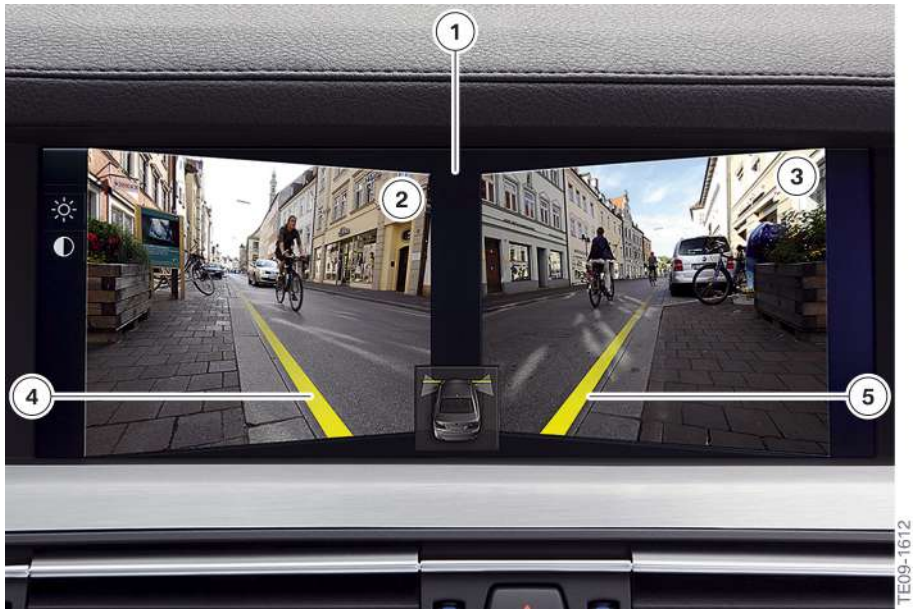
3.5.2. Side View

ASSISTANCE SYSTEMS.
3. PARKING.

Function

The Side View is realised by means of 2 digital cameras which are located in the front wheel arches (left/right) in each case. The function supports the driver when driving into streets and intersections where the side view is obscured.

Side View was used for the first time in the F01/F02.



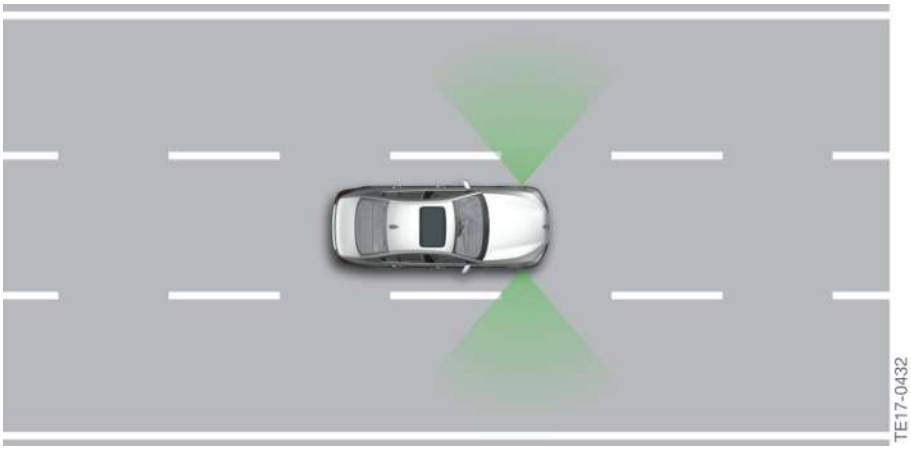
Side View display in the central information display CID

Index	Explanation
1	Side View display in split view
2	Display of left Side View camera
3	Display of right Side View camera
4	Projected front of vehicle in the left view
5	Projected front of vehicle in the right view

Sensors

2 cameras are installed in the front wheel arches in each case for the Side View function.

ASSISTANCE SYSTEMS.
3. PARKING.



Side View sensors

Operation

Side View is activated by pressing the camera system button.



Camera system button

Index	Explanation
1	Camera system button

ASSISTANCE SYSTEMS.
3. PARKING.

3.5.3. Panorama View

Panorama View is the further development of the previous Side View and was used for the first time in the F15. The Panorama View function is not realised by means of 2 cameras installed at the side of the vehicle as previously. The image information is supplied by a camera installed between the radiator grilles.

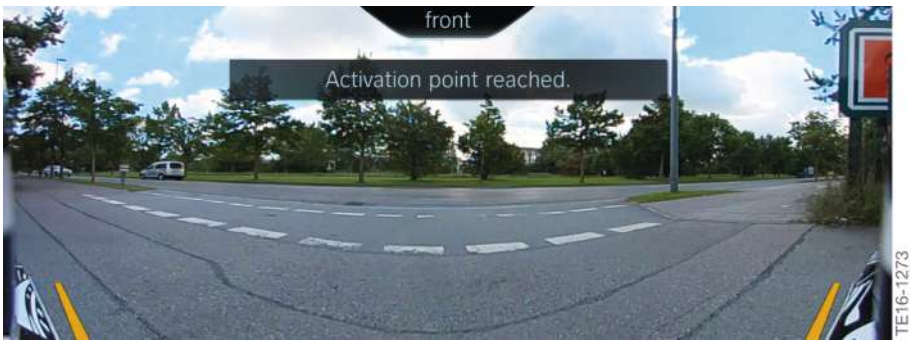
Depending on the gear engaged, a panorama view of the front or rear vehicle area is shown on the CID.



Panorama View display on the CID

Automatic activation point

Positions at which the Panorama View should be automatically switched on can be stored as activation points. A prerequisite for this is GPS reception. Up to 10 activation points can be stored. The function was offered for the first time in the G11/G12.



automatic switch on at a set activation point

Storing an activation point

In order to store an activation point, it is necessary to stop briefly and activate Panorama View at the point where Panorama View is to be activated automatically. Select the menu option “Add activation point”.

Panorama View will then be automatically activated at the stored location.



Storing an activation point

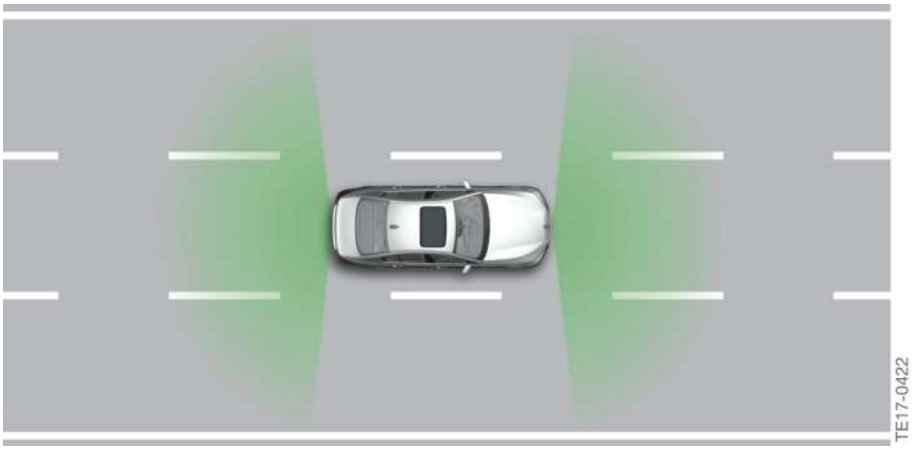
Index	Explanation
1	Add activation point

If possible, the activation points are saved with the town/city and street or with the GPS coordinates. The activation points are stored in relation the direction of travel.

Sensors

One camera in each case is installed at the front and rear for Panorama View.

ASSISTANCE SYSTEMS.
3. PARKING.



Panorama View sensors

Operation

Panorama View either activates itself or can be activated by pressing the camera system button



Camera system button

Index	Explanation
1	Camera system button

ASSISTANCE SYSTEMS.
3. PARKING.

3.5.4. Top View

The driver can see a bird's eye view of his vehicle on the central information display CID with the help of the rear view camera and 2 cameras in the exterior mirrors. This provides assistance when manoeuvring or parking.



Exterior mirror camera

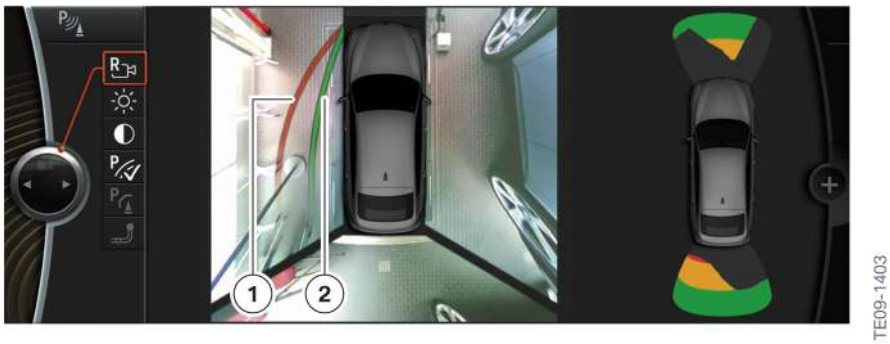
Index	Explanation
1	Exterior mirror camera

ASSISTANCE SYSTEMS.
3. PARKING.



Top View

Index	Explanation
1	Obstruction
2	Warning about the obstacle



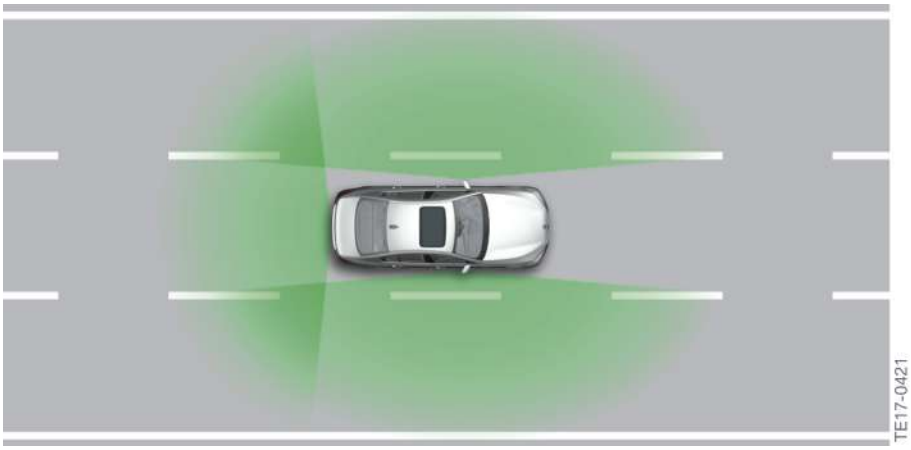
Top View perspective with driving line and turning circle line

Index	Explanation
1	Turning circle line
2	Lane line

When an obstacle is detected, the driver receives a visual and acoustic warning in the CID with the assistance of the Park Distance Control PDC function.

Sensors

2 top view cameras in the exterior mirrors and a rear view camera are needed for the function.



Top View sensors

Operation

Top View is activated by pressing the camera system button.

ASSISTANCE SYSTEMS.
3. PARKING.



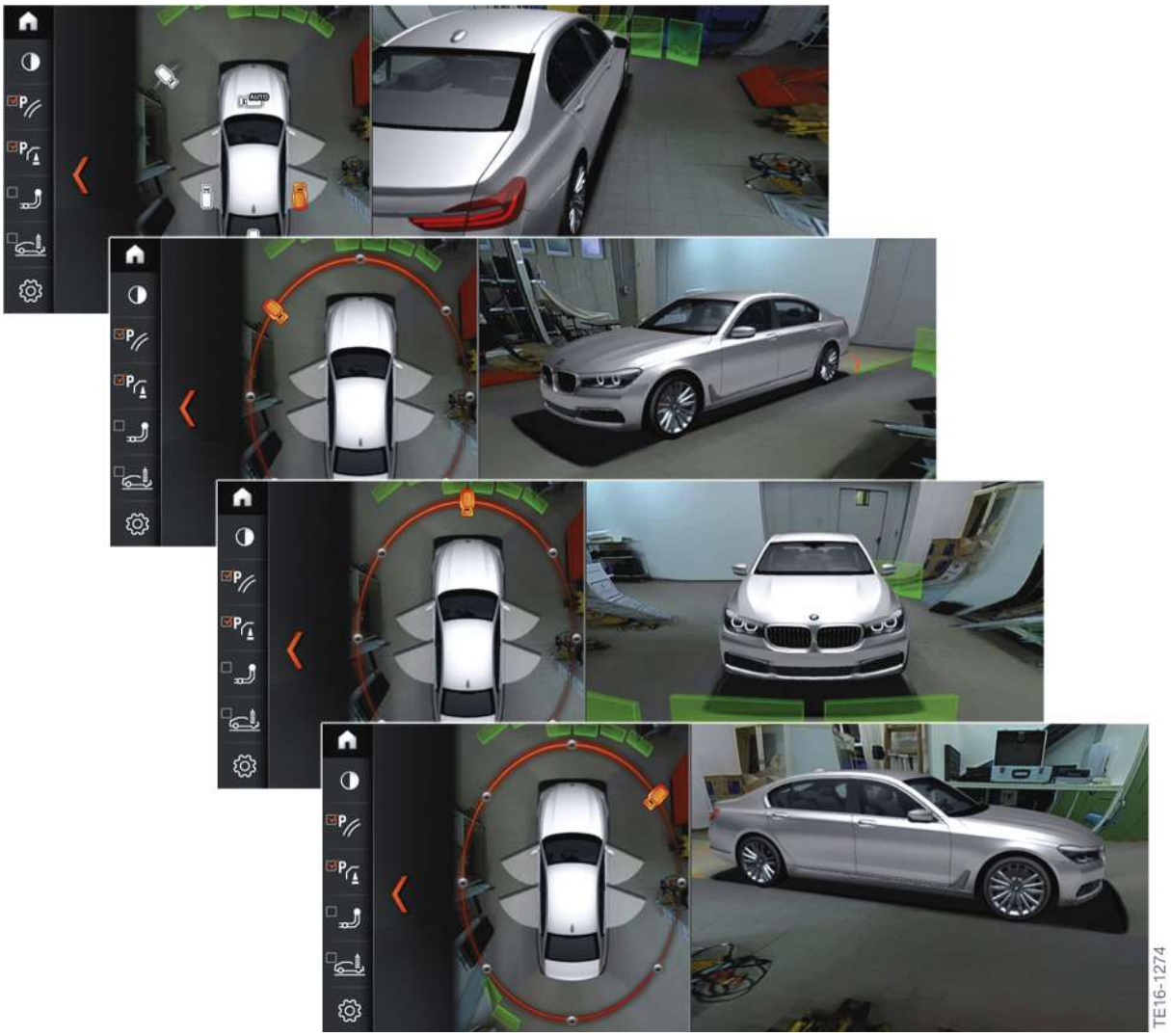
Camera system button

Index	Explanation
1	Camera system button

3.5.5. Surround view

Surround View includes the rear view camera, Panorama View and Top View. The images from the individual cameras are combined into a spatial view around the vehicle from different perspectives using 3D computer graphics.

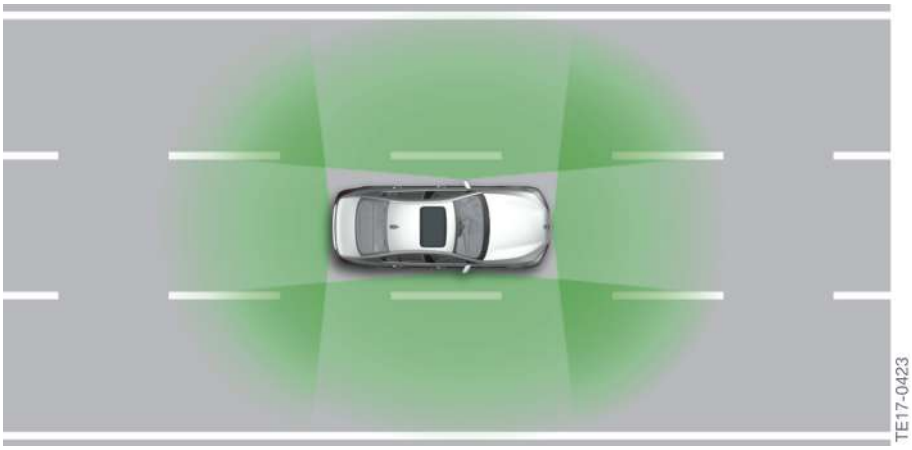
ASSISTANCE SYSTEMS.
3. PARKING.



Examples of camera views for Surround View

Sensors

4 cameras are installed for the Surround View function: 1 front camera, 1 rear view camera and 2 top view cameras in the exterior mirrors.



Surround View sensors

Operation

Surround View is activated by pressing the camera system button.



Camera system button

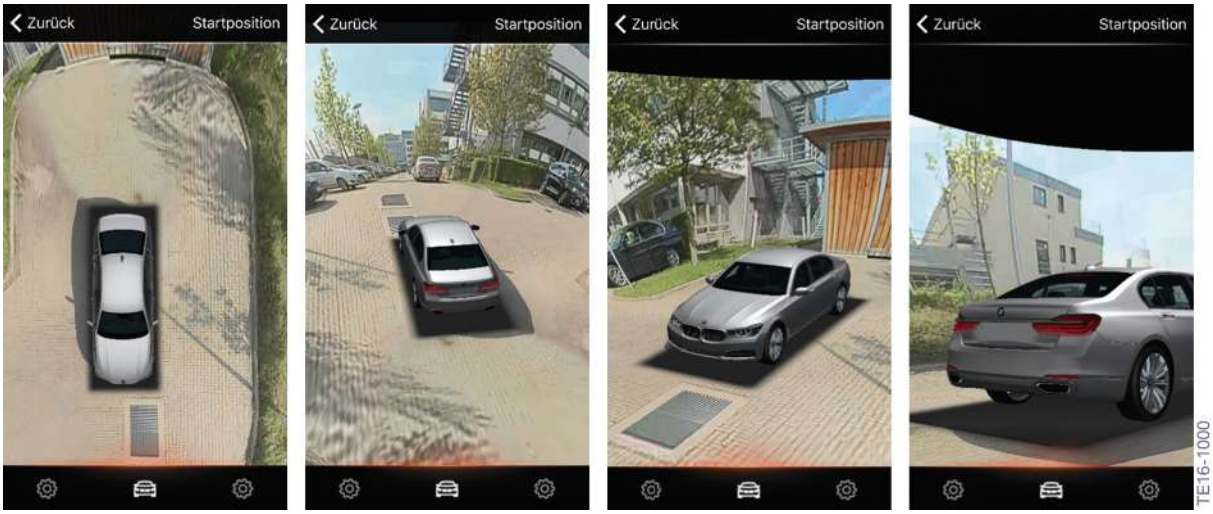
Index	Explanation
1	Camera system button

3.5.6. Remote 3D View

Remote 3D View was used for the first time in the G30. The function allows the customer to view images of his parked vehicle on a smartphone.

ASSISTANCE SYSTEMS.
3. PARKING.

Remote 3D View is controlled via the BMW Connected app. The vehicle must have the optional equipment “Surround View”.



Remote View 3D display on smartphone

Functional principle

The following graphic explains the functional principle of Remote 3D View:



Functional principle of Remote 3D View

Index	Explanation
1	Request to the BMW back end via the BMW Connected app
2	Request from the BMW back end to the vehicle
3	The vehicle sends the images from the Surround View cameras to the BMW back end (4 individual images)
4	The BMW back end sends the images to the smartphone. In the smartphone, the BMW Connected app combines the 4 individual images into a 360° panorama view.

ASSISTANCE SYSTEMS.
3. PARKING.

Images

The following graphic shows an example of the 4 individual images. These 4 images are combined into one graphic via the BMW Connected app.



Example of 4 individual images

3.6. MINI

The MINI models are equipped with the following assistance systems from the "Parking" area:

- Park Distance Control (PDC)
- Auto PDC
- Parking manoeuvring assistant PMA
- Reversing camera.

ASSISTANCE SYSTEMS.
4. DRIVING.

4.1. Lane departure warning

The lane departure warning supports drivers by warning them if they are unintentionally straying from the lane.

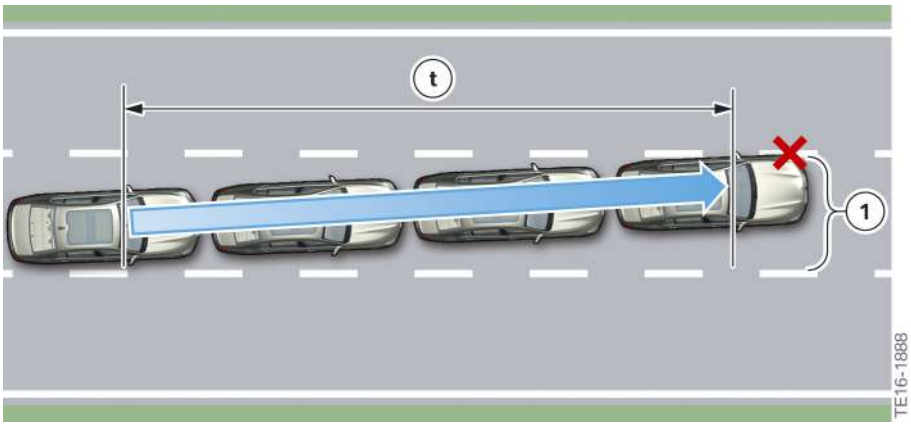
The lane departure warning function has been used at BMW since March 2007.

4.1.1. Function

The lane departure warning function detects lane markings from a speed of approximately 70 km/h and warns the driver against leaving the lane by means of steering wheel vibrations.

The lane departure warning function is deactivated if the turn indicator is activated. This prevents warnings being issued for an intentional lane change.

The system issues a warning when the vehicle moves toward a detected lane marking. For this purpose, the time is calculated until it can be expected that the vehicle will cross the lane marking. The warning is activated in good time before the vehicle crosses the lane marking.

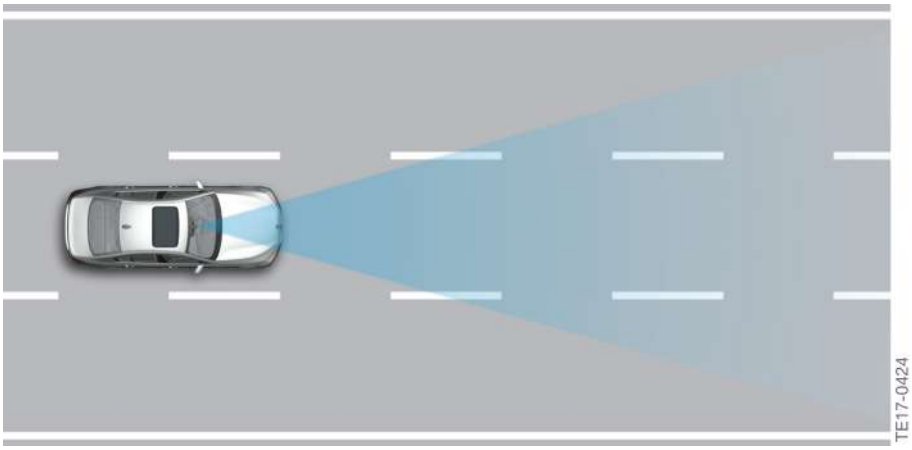


Principle of lane departure warning function

Index	Explanation
1	Current lane
t	Calculated time until vehicle crosses the lane marking

4.1.2. Sensors

The lane markings are detected by the KAFAS camera.



Sensors for lane departure warning function

4.1.3. Operation

The lane departure warning function is activated either by means of the intelligent safety button or by a separate button (vehicle-dependent).



Operation of lane departure warning function

Index	Explanation
A	Operating facility for assist systems
B	Intelligent Safety button

4.1.4. Display

When the lane departure warning function is activated, this is indicated on the instrument cluster by yellow bars which represent the road. The arrows indicate where a lane marking has been detected by the KAFAS camera. In this example, the KAFAS camera detects a lane marking on both sides.

ASSISTANCE SYSTEMS.
4. DRIVING.



Display of lane departure warning function on instrument cluster

Display from G11

In the G11, the lane departure warning function is represented by a new symbol in the instrument cluster:

Display	Explanation
	Lane departure warning activated. Prerequisites for warnings are no longer met.
	Lane departure warning activated. At least one lane marking is detected.

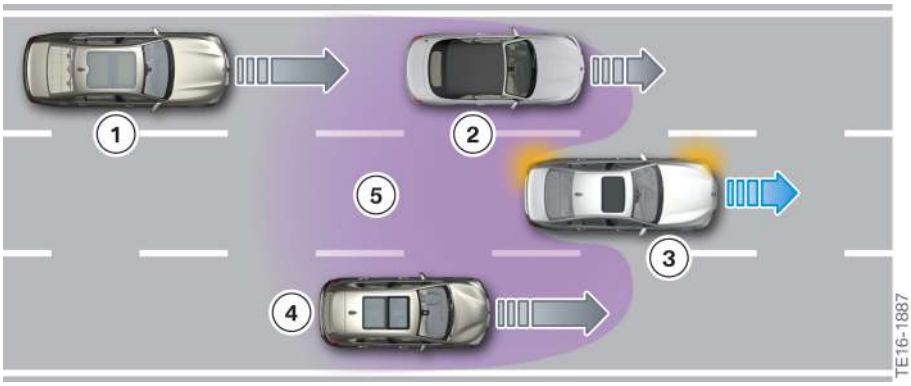
4.2. Lane change warning

4.2.1. Function

The lane change warning system SWW can detect traffic situations that could pose a risk if the driver changes lane. These traffic situations include vehicles approaching quickly from behind or vehicles in the driver's blind spot.

The lane change warning function was offered for the first time in the F01/F02.

2 radars sensors in the rear area of the vehicle are used to detect vehicles approaching from behind. The lane change warning function must be activated in the vehicle. The function is active as from a speed of approx. 30 km/h.

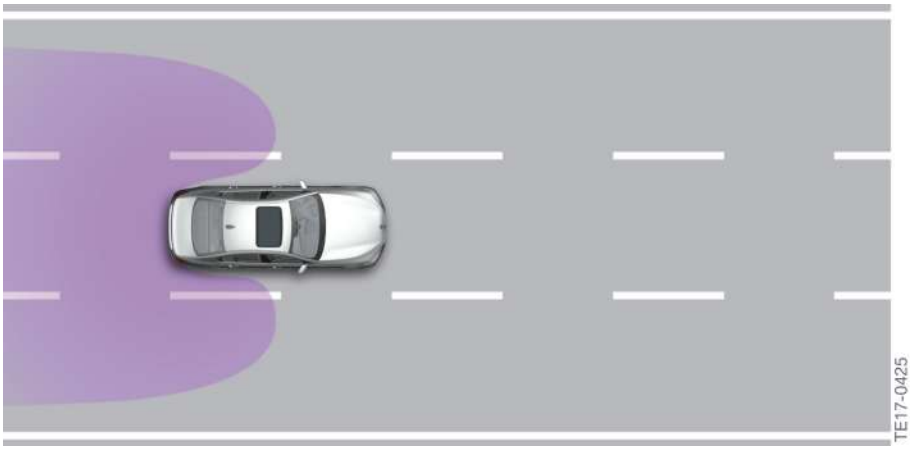


Lane change warning

Index	Explanation
1	Approaching vehicle in the left adjacent lane
2	Vehicle driving at the same speed in the left adjacent lane
3	Own vehicle planning to change lane to the left
4	Faster vehicle in the right adjacent lane
5	“Lane change area”

4.2.2. Sensors

Vehicles are detected by means of 2 radar sensors in the rear bumper. The range of the sensors is between 50 m and 70 m.



Lane change warning sensors

Both radar sensors are visually identical. In functional terms, the rear right radar sensor is the master. The rear left sensor is the slave. Both radar sensors have diagnostic capability.



Radar sensors for lane change warning function

4.2.3. Informing and warning

Information

When a vehicle is detected and the system is activated, the driver is informed of the situation by an unobtrusive indicator in the exterior mirror. By having this information before making a lane change manoeuvre, the driver can confidently prepare for the lane change and avoid critical situations from the outset.



LED in exterior rearview mirror

Warning

The warning must be much more conspicuous that an information display. It must warn the driver quickly and directly if he is intended to change lane in spite of a dangerous situation. A warning is output if the following conditions are met:

- Conditions for information met
- Turn indicator active on the side for warning.

The visual warning is issued by flashing of the warning light in the exterior rearview mirror. The steering wheel also vibrates. The driver is therefore warned both visually and haptically.



Warning of lane change warning function

Index	Explanation
1	Left turn indicator activated
2	Warning light in left exterior rearview mirror flashes with high intensity
3	Warning light in left exterior rearview mirror lights up with low intensity
4	Steering wheel vibrates



If the driver changes lane without using the turn indicator, he is provided with only a discreet information display by the lane change warning function. A warning is issued only in combination with the turn indicator.

4.2.4. Rear-end collision prevention

The radar sensors can also detect situations where there is an imminent risk of a rear-end collision. The following preventive functions are then activated on the vehicle:

- Hazard warning lights system
- Elimination of belt slack
- Backrests are moved to upright position
- Side windows are closed.

4.3. Side collision warning

4.3.1. Function

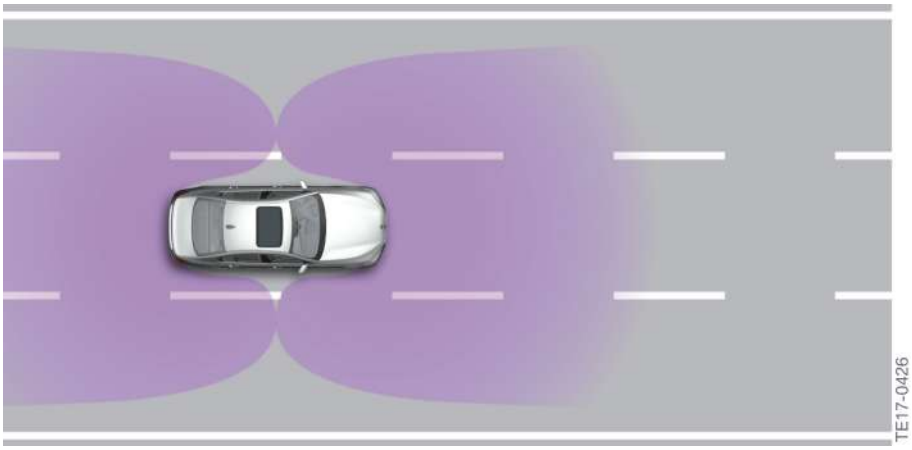
The side collision warning assists the driver in avoiding a potential side collision. If the vehicle detects an object such as a vehicle or crash barrier in the side area where there is a risk of a potential side collision, the system assists the driver in avoiding the collision.

The driver is warned by means of LEDs in the exterior rearview mirror and steering wheel vibrations. There is also an active steering intervention which additionally moves the vehicle to a safe area in its own lane. The steering intervention is initiated within a speed range of between 70 km/h and 210 km/h.

The side collision warning is part of the lane departure warning system with side collision protection.

4.3.2. Sensors

Four radar sensors monitor the area next to the vehicle. The rear vehicle area is monitored by the radar sensors of the lane change warning function. The front vehicle area is monitored by 2 additional radar sensors.



Side collision warning sensors

4.3.3. Operation

The side collision warning is activated or deactivated and personalised by means of the intelligent safety button.



Side collision warning intelligent safety button

4.4. Lane keeping assistant with active side collision protection

The lane keeping assistant with active side collision protection was used for the first time in the G11/ G12.

4.4.1. Function

The system assists the driver by means of corrective steering interventions in the speed range between 70 km/h and 210 km/h. The vehicle is kept in lane and potential side collisions are actively avoided. The system endeavours to orient the vehicle in its lane according to the situation and within the framework of the system limits.

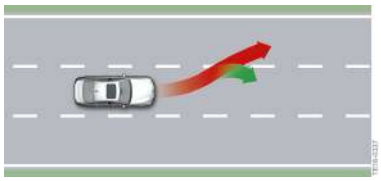
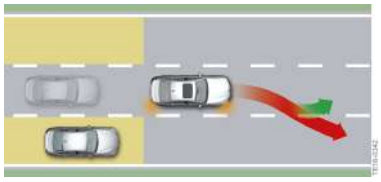
The system consists of the following four subfunctions:

- Side collision warning
- Reduced side collision warning
- Lane departure warning
- Lane Change Warning.

4.4.2. Overview

The following table provides an overview of the function variants depending on situation:

Speed range	Warning	Steering interventions
Side collision warning		
70 km/h – 210 km/h	• Issued if the vehicle detects that it is getting critically close to another vehicle to the side • Steering wheel vibration • Warning light in exterior rearview mirror	• In the direction of the free traffic space Requirements: • Detected lanes • Sufficient available space on the roadway 
Reduced side collision warning		
30 km/h – 210 km/h	• Issued if the vehicle detects that it is getting critically close to another vehicle to the side • Steering wheel vibration • One-off steering pulse at 30 km/h-75 km/h • Warning light in exterior rearview mirror	• No steering intervention but only a one-off steering pulse in the direction away from the collision-critical object.
Lane departure warning		

Speed range	Warning	Steering interventions
70 km/h – 210 km/h	• Triggering when a lane departure situation is detected • Steering wheel vibration • Warning light in exterior rearview mirror	• In direction of own lane Required: • Detected lanes 
Lane change warning		
70 km/h – 210 km/h	• Triggering in the event of a detected collision-critical lane change • Steering wheel vibration • Warning light in exterior rearview mirror	• In the direction of the free traffic space Requirements: • Detected lanes • Sufficient available space on the roadway 

4.5. Cruise control with braking function

The cruise control with braking function (Dynamic Cruise Control) has been used at BMW since the BMW 3 Series (E9x). It relieves the burden on the driver on quiet roads by maintaining a constant speed regardless of the resistance to vehicle motion (gradient, payload).

4.5.1. Operation

Cruise control with braking function is operated via the multifunction steering wheel.



Operation of cruise control with braking function

4.6. Speed limit

The driving speed can be restricted to a value between 30 km/h and 250 km/h via the multifunction steering wheel. The set speed is then not exceeded unless the accelerator pedal is fully depressed.

4.6.1. Operation

The speed limit is operated by means of the LIM button on the multifunction steering wheel.



Operation of cruise control with braking function

4.6.2. Display

An activated speed limit is indicated by an indicator light in the instrument cluster.



Speed limit display

Index	Explanation
1	Indicator light for speed limit LIM
2	Set speed limit

4.6.3. Speed Limit Assist

The speed limit function has been extended by the Speed Limit Assist function since the F48. This allows the driver to adopt the current road speed limit as the new vehicle speed limit.

The current road speed limit is detected by the road sign detection function and displayed in the instrument cluster. If the currently detected road speed limit deviates from the last detected limit, a speed tip is displayed for around 20 s.

The suggestion can be adopted as the new vehicle speed limit by means of the rocker button on the multifunction steering wheel.

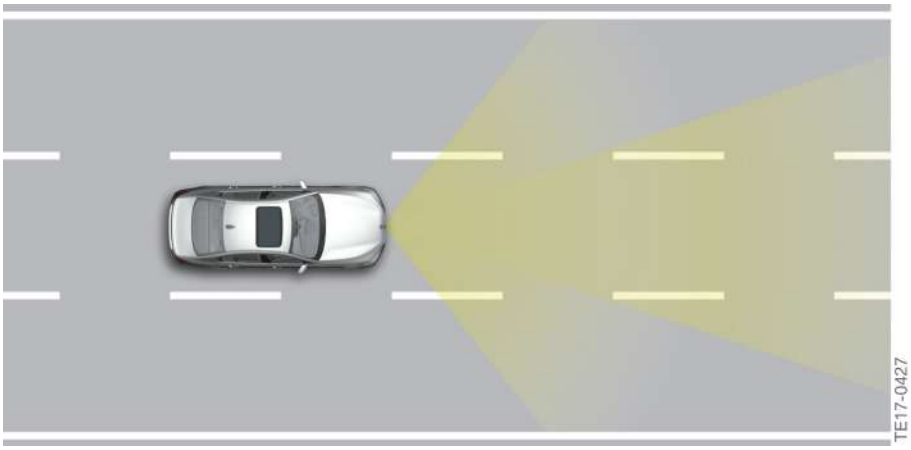
4.7. Active speed control

4.7.1. Function

Active cruise control with Stop&Go function maintains a constant distance from the vehicle in front. For example, the system brakes the vehicle to standstill in a traffic jam situation, and then drives off again automatically as soon as it is possible to continue.

4.7.2. Sensor

A radar sensor installed at the front of the vehicle detects the vehicle in front. On the basis of the measured data, a preset distance from the vehicle in front is maintained.



Radar sensor for active cruise control

4.7.3. Operation

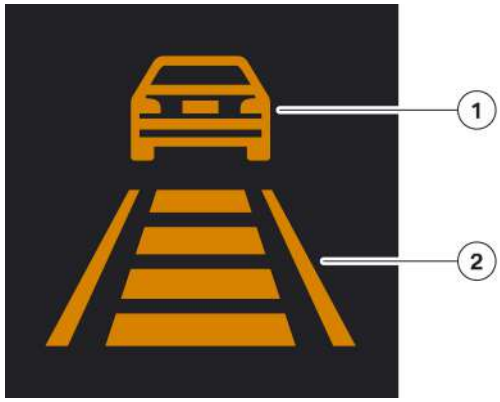
The active cruise control function is operated by means of buttons on the multifunction steering wheel.



Operation of active cruise control on multifunction steering wheel

4.7.4. Display

The set distance and the detected vehicle in front are displayed on the instrument cluster or, where applicable, on the Head-Up Display HUD.



Active cruise control display on instrument cluster

Index	Explanation
1	Vehicle driving ahead detected
2	The bars show the selected distance from the vehicle in front.

4.7.5. Overview of generations

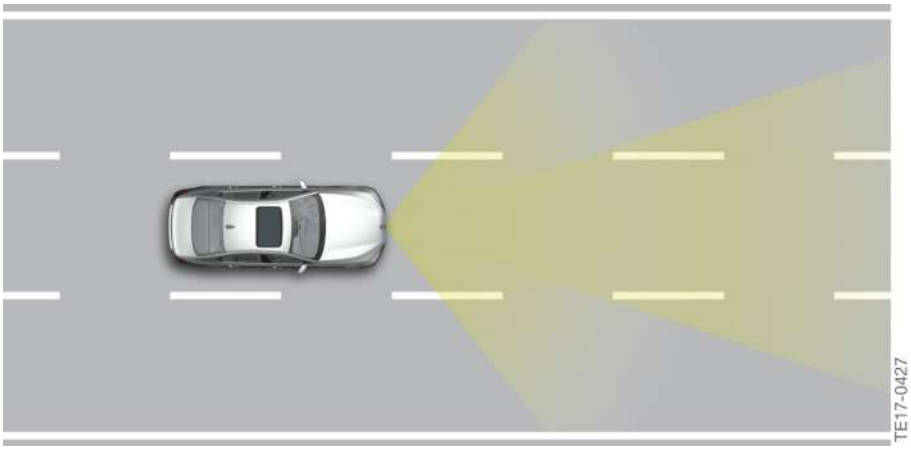
The active cruise control functions have been continuously further developed since the system was introduced in the E60. At the start of active cruise control, the system was able to maintain the distance from the vehicle driving in front. The system was active in a speed range between 30 km/h and 180 km/h.

F01/F02

The F01/F02 has several radar sensors. The visible radar sensor at the front is responsible for the far range and 2 radar sensors behind the front bumper are responsible for the near range. The two near range radar sensors are required for the Stop&Go function. Stop&Go means that the system can brake the vehicle to a standstill. The system can drive off again automatically in a period of approx. 3 s.

F10

The two radar sensors for the near range have been omitted since the F10. A common radar sensor is installed for the near and far ranges.

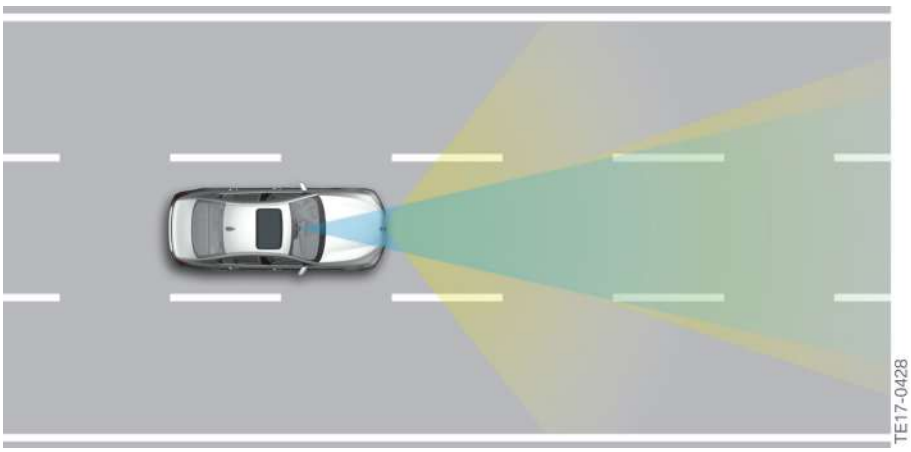


Radar sensorF10

F01/F02 LCI

Previously, only a radar sensor was used for active cruise control. With the F01 LCI, the sensor system for active cruise control was extended by the KAFAS camera.

This makes it possible to also detect stationary vehicles. The fusion of the radar sensor and KAFAS camera allows clear identification of lane markings and makes it possible to distinguish between stationary vehicles and other fixed objects. In addition, the adjustment range was increased from 180 km/h to 210 km/h.

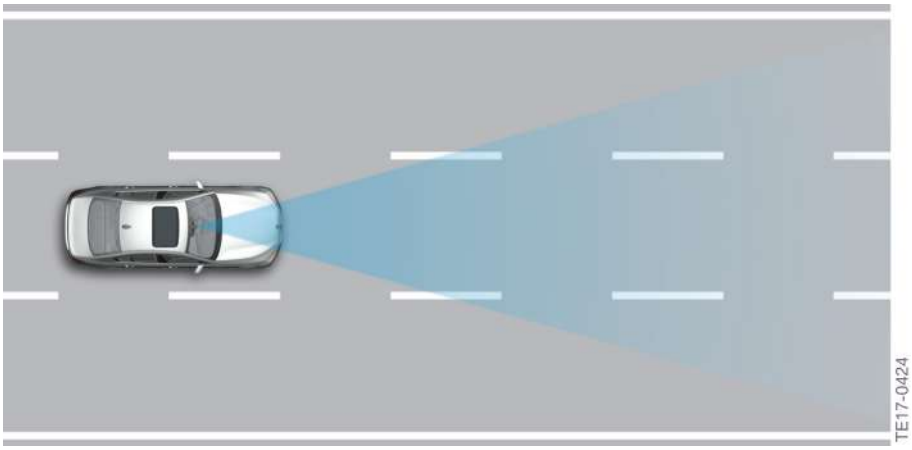


Sensor system for active cruise control with radar and KAFAS camera

I01, F45/F46 and MINI from F56

These vehicles have camera-based cruise control. The functionality, operation and displays are the same as with the radar sensor. Here, the vehicle driving in front is detected only via the KAFAS camera. The KAFAS camera can detect vehicles ahead at a distance of up to 90 m. The front camera detects stationary vehicles and vehicles that are swinging in or out, and thus makes anticipatory control possible. Since the KAFAS camera is an optical sensor, the quality of object detection depends on the weather conditions. The maximum control speed is limited to 140 km/h with the camera-based system.

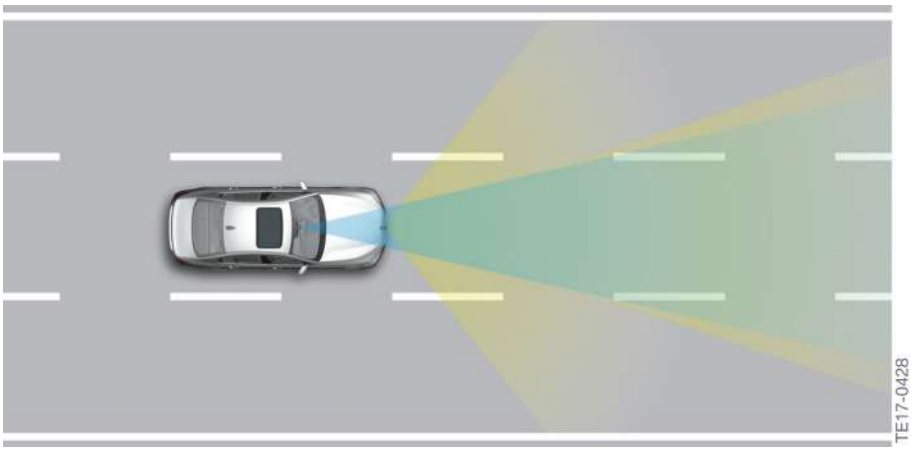
ASSISTANCE SYSTEMS.
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Sensor system for active cruise control with KAFAS camera

G11/G12

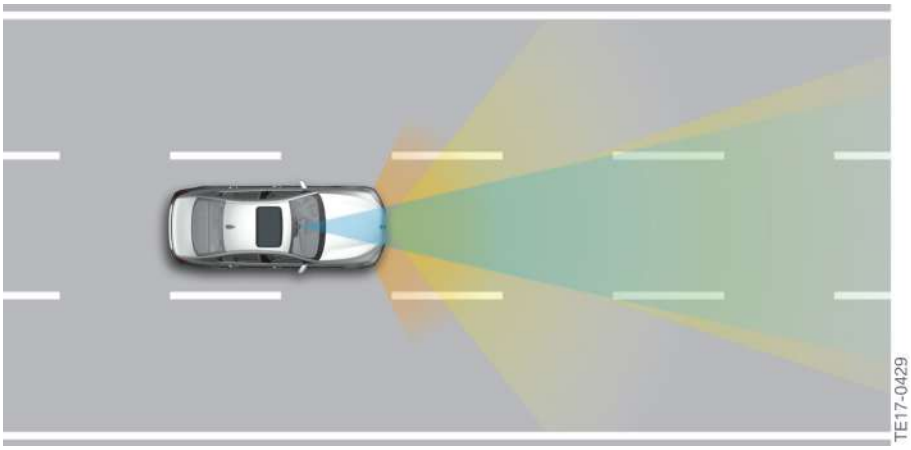
The G11/G12 is able to more clearly detect vehicles and objects with the KAFAS stereo camera. This contributes to smoother following behaviour.



Fusion of radar sensor with KAFAS stereo camera

G30

The time for automatically driving off again was increased from 3 s to 30 s. The extended automatic driving off time is possible only on motorways or structurally separated roads. The ultrasonic sensors of Park Distance Control PDC are additionally used for this function.



Sensors for active cruise control in the G30

The radar sensor in the G30 can also include the vehicle in front of the vehicle directly in front in calculations. The advantages of this are as follows:

- Improved "driver-like" behaviour
- Smoother following behaviour
- Improved automatic driving off behaviour
- Improved behaviour when approaching objects.

The radar sensor is heated to help prevent weather-related failure of the system.

4.7.6. Overview

The following table provides an overview of the different generations:

Generation	Sensors	Special features
From F01/F02	Radar sensors for near range Radar sensor for far range	
From F10	Radar sensor for far and near ranges	1 common radar sensor
From F01/F02 LCI	Radar sensor + KAFAS camera	The fusion of the systems permits detection of stationary vehicles. Detection of vehicles in adjacent lanes Increase on speed to 210 km/h

Generation	Sensors	Special features
From I01, MINI F56 and F45/ F46	KAFAS camera	Camera-based cruise control
From G11/G12	Radar sensor + KAFAS camera	Improved detection through use of a KAFAS stereo camera
From G30	Radar sensor + KAFAS camera + ultrasonic sensors	Time for automatically driving off again increased from 3 s to 30 s. Radar sensor is heated

The following table contains an overview of the sensors that are used in the vehicles for active cruise control:

Vehicle	Sensor
F45, F46 F48 F54, F55, F56, F57, F60	KAFAS camera
F20, F21 F22, F23 F30, F31, F34 F32, F33, F36 (F01 and F10 up to life cycle impulse)	ACC sensor
F1x F0x F15, F16 F25, F26 G11, G30, G31	ACC sensor + KAFAS camera

4.7.7. Speed Limit Assist

The Speed Limit Assist function is a sub-function of active cruise control. If the vehicle detects a road speed limit, this is displayed in the instrument cluster. The driver can accept this limit. This is done by pressing the rocker button on the multifunction steering wheel.

Speed Limit Assist was offered for the first time in the G11/G12.

If active cruise control is activated and Speed Limit Assist switched on, this is shown in the instrument cluster in the form of a green symbol and the word “ASSIST”.



Speed Limit Assist display in instrument cluster

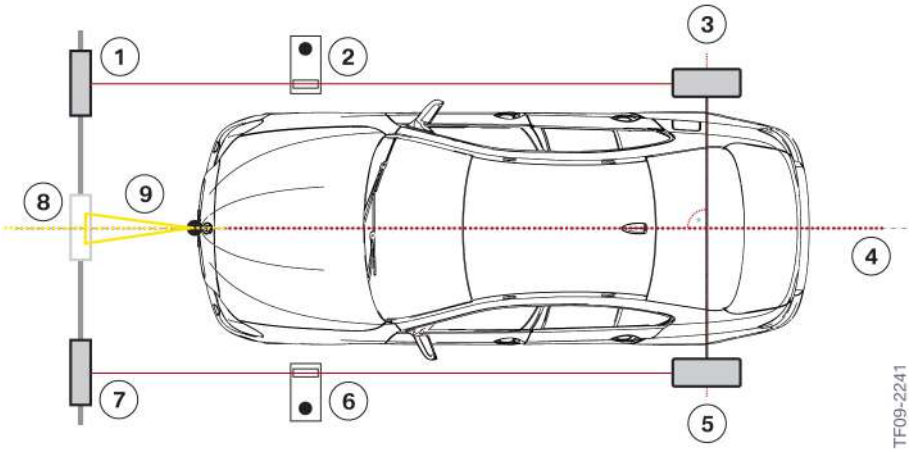
Index	Explanation
1	The range between the current speed and the new set speed is marked in green in the speed indicator.
2	Display lights up when system is switched on
3	A green arrow shows whether the new speed is higher or lower than the set speed.
4	Road sign recognition display
5	Road sign lights up green: change detected to a road speed limit with immediate effect.
6	The remaining distance display indicates a change to the speed limit after a certain distance.

4.7.8. Alignment of radar sensor

The radar sensor must be adjusted in the following situations:

- Replacement of radar sensor
- Adjusting the chassis and suspension
- Corresponding fault memory entry.

Alignment of the radar sensor is performed via test module and special tool.



Adjustment of radar sensor

Index	Explanation
1	Mirror, right
2	Slot aperture, right
3	Laser pointer, right
4	Longitudinal axis of the vehicle
5	Laser pointer, left
6	Slot aperture, left
7	Mirror, left
8	Adjusting mirror
9	Radar sensor

4.8. Traffic Jam Assist

4.8.1. Function

In traffic jam situations, the traffic jam assistant controls the speed and keeps the vehicle in lane through supporting steering interventions. The driver must hold the steering wheel with at least one hand during the active control of the traffic jam assistant. This is detected by a special sensor in the steering wheel. Within the framework of the given possibilities, the traffic jam assistant adapts the speed to a vehicle ahead.

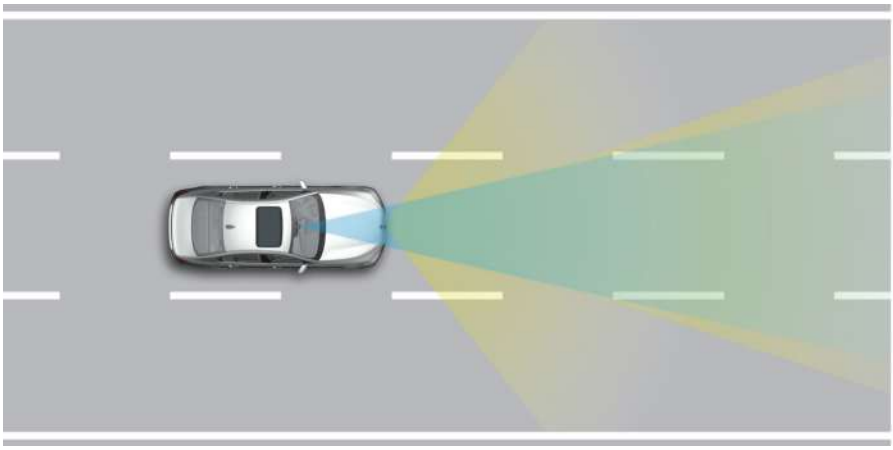
The traffic jam assistant is a further development of the active cruise control system. The difference is that active cruise control performs only longitudinal control (acceleration, deceleration). The traffic jam assistant also performs lateral control (steering interventions) in addition to longitudinal control.

Longitudinal control functions in a similar way to active cruise control. The driver selects a set speed and a distance to the vehicle ahead.

The left and right lane markings are detected by the KAFAS camera. For lateral control with assisted steering, a vehicle ahead must also be detected, in addition to the detected lane markings. If the corresponding electronics detects contact with the steering wheel, the traffic jam assistant supports the driver with specific steering assistance.

4.8.2. Sensors

The traffic jam assistant uses the same sensors that are required for active cruise control.

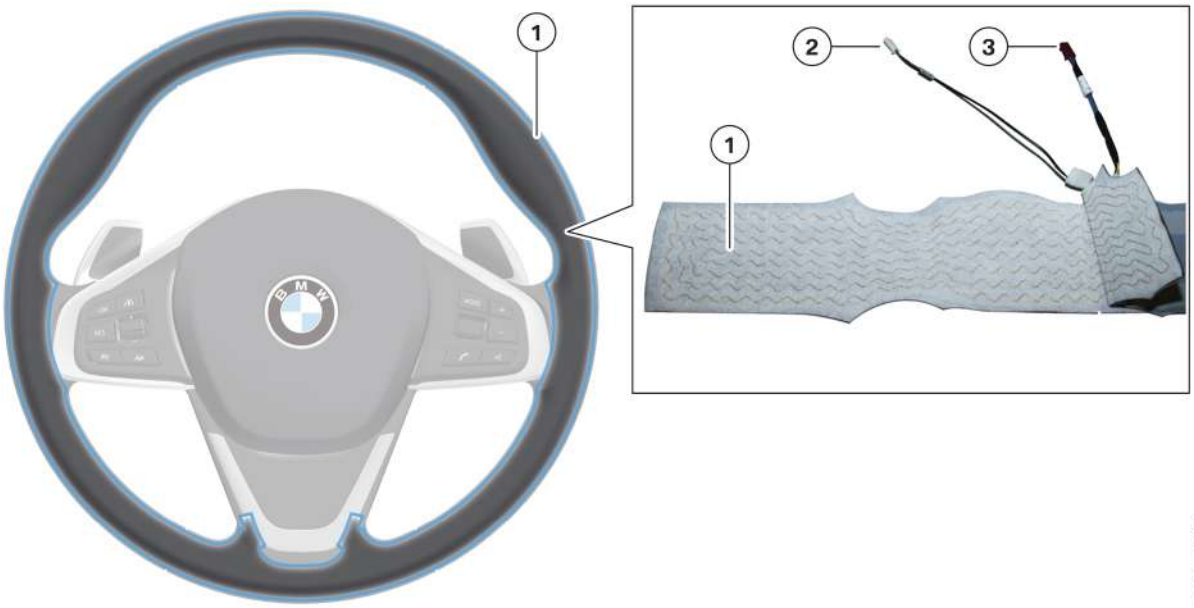


Traffic jam assistant sensors

If only the KAFAS camera is used as a sensor for active cruise control, only the KAFAS camera is also used for the traffic jam assistant.

Touch detection

In order to use the traffic jam assistant, at least one hand must be touching the steering wheel. A sensor mat is fitted on the steering wheel for hand detection.



Sensor mat for touch detection

Index	Explanation
1	Sensor mat
2	Connection for capacitive touch detection
3	Connection for steering wheel heating

The evaluation electronics for the touch detection is in the steering wheel and cannot be replaced separately for Service.



Evaluation electronics for touch detection

4.8.3. Operation

The traffic jam assistant is activated via the traffic jam assistant button on the multifunction steering wheel.



Traffic jam assistant button on multifunction steering wheel

4.8.4. Displays

The following displays are shown in the instrument cluster:

Display	Explanation
	Traffic jam assistant deactivated. ACC Stop & Go controls with reference to the distance set.
	Traffic jam assistant and ACC Stop&Go on standby.
	Traffic jam assistant on standby. ACC Stop&Go controls with set distance.

Display	Explanation
	Traffic jam assistant activated. The system controls the speed and assists in keeping the vehicle in lane.
	Rolling bar during the journey: Speed is no longer increased by the traffic jam assistant at 60 km/h. If the speed is increased by the driver, the system no longer provides steering assistance.
	Red flashing and acoustic signal: traffic jam assistant has been interrupted. The system does not steer automatically. ACC Stop&Go adjusts.

4.8.5. Overview

The following table contains on overview of the different generations:

Vehicle/Generation	Functional characteristics
I01 1st generation	• 0 km/h to 40 km/h • Motorways and single-carriageway roads with central reservations
F45 1st generation with function enhancement	• 0 km/h to 60 km/h • Motorways and single-carriageway roads with central reservations
G11/G12 2nd generation	• 0 km/h to 70 km/h • No road-type restrictions

4.9. Steering and lane control assistant

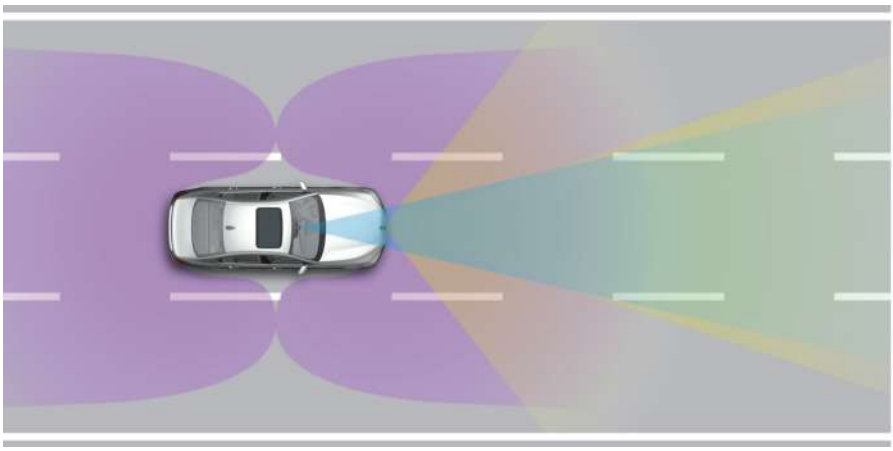
The steering and lane control assistant functions in a similar way to the traffic jam assistant. The system assists the driver in keeping the vehicle in lane. For this purpose, the system independently performs steering, e.g. when cornering. The steering and lane control assistant has been offered since the G11/G12.

4.9.1. Function

Depending on the speed, the system is guided by the lane edges or vehicles driving ahead. The steering wheel must be touched by at least one hand. The steering and lane control assistant is active in the speed range between 70 km/h and 210 km/h. The traffic jam assistant supports steering and lane control up to a speed of 70 km/h.

4.9.2. Sensors

The system determines the position of the lane markings and vehicle driving in front by means of the radar sensors and KAFAS camera.



Sensors of steering and lane control assistant

4.9.3. Operation

The system is operated by means of the traffic jam assistant button.



Traffic jam assistant button on multifunction steering wheel

4.10. Tailgate warning

4.10.1. Function

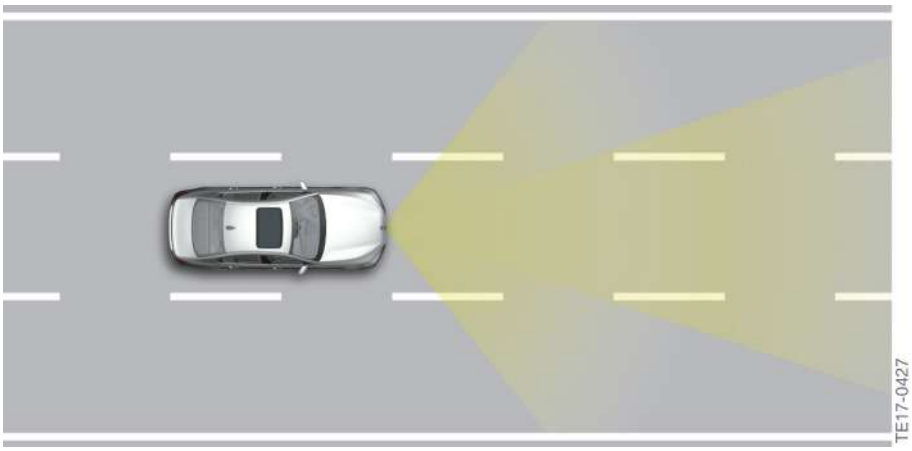
If the vehicle ahead brakes sharply, a visual and acoustic warning is output in order to help prevent a potential imminent collision. The warning is output so that a collision can be avoided if braking is initiated immediately.

The radar sensor detects a potential imminent collision and warns the driver by a warning light in the instrument cluster and, where applicable, in the Head-Up Display. At the same time, the brake system is prefilled and the thresholds of the hydraulic brake assistant reduced.

Since the F01/F02, the adaptive brake assistant has extended to include a collision warning function. The collision warning is divided into an advance warning and an acute warning. The collision warning functions via a radar sensor in the F01/F02, but is independent of the active cruise control function.

The collision warning warns the driver with an advance warning and acute warning. If an advance warning is output, strong braking is sufficient to avoid a collision. In the case of an acute warning, the driver must brake immediately with maximum braking force in order to avoid a collision.

4.10.2. Sensors



Radar sensor for active cruise control

Collision warning via KAFAS camera

Since the F20, the collision warning has also been realised via the KAFAS camera. The functions are unchanged compared with a radar sensor. There are 2 warning stages (advance warning and acute warning). Vehicles that have radar-based active cruise control as optional equipment use the sensors that are installed in the vehicle for the collision warning function. If the vehicle has active cruise control, for example, the collision warning is realised using the radar sensor. If the vehicle does not have active cruise control but has a KAFAS camera, the collision warning is realised using the KAFAS camera.

4.10.3. Operation

The collision warning function is operated either by means of the intelligent safety button or by means of the operating facility for the assistance systems



Operation of collision warning function

Index	Explanation
A	Intelligent safety operating facility
B	Intelligent Safety button

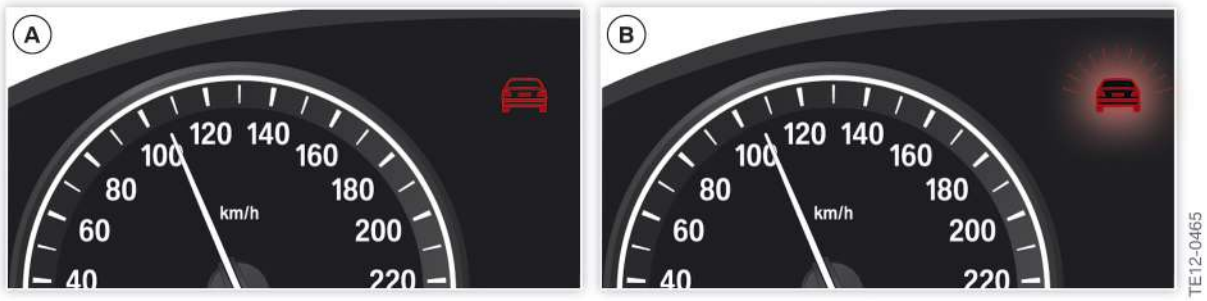
The threshold value for the advance warning can be adjusted:

- Early
- Late
- Off.

If “Off” is selected, no advance warning will be output. In this case, the acute warning is output directly.

4.10.4. Displays

The collision warning is displayed in the instrument cluster or, where applicable, in the Head-Up Display HUD.



Warning light for collision warning

Index	Explanation
A	Advance warning (visual)
B	Acute warning (visual and acoustic)

4.10.5. Collisiont warning with brake activation

Since the F10 , the collision warning function has been extended by a brake activation function.

As from a speed of approx. 15 km/h, the system warns about a possible collision in 2 stages (advance warning and acute warning). Stationary or moving objects are taken into account here if they are in the detection range of the radar sensor. Brake intervention takes place only in the event of an acute warning. The brake intervention is limited in time.

Due to the fusion of the radar sensor and KAFAS camera, it was possible to increase the deceleration of the brake activation function as from theF01/F02 LCI.

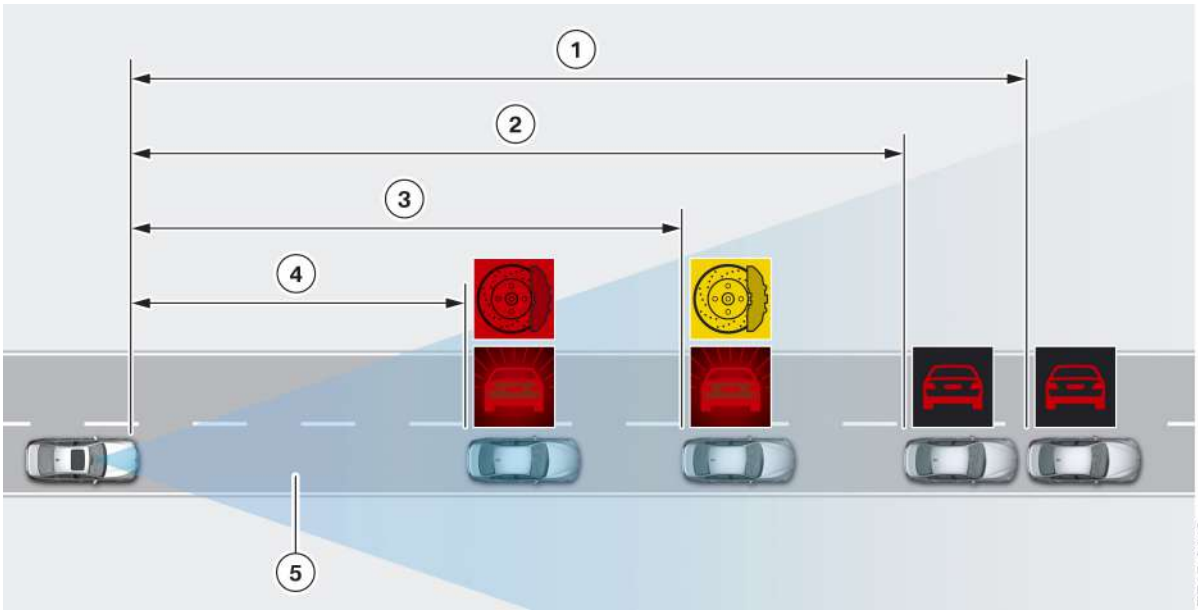
A brake intervention takes place at speeds up to approx. 50 km/h. Stronger braking takes place if when the object has been verified by the KAFAS camera. The vehicle can be braked to a standstill.

A brake intervention also takes place at speeds above 50 km/h. However, this is not as strong as in the lower speed range.

The brake intervention is limited to 10 km/h at speeds above 210 km/h.

4.10.6. Camera-based collision warning

2 additional collision warning functions were introduced at BMW with the I01, F45 or F56: collision warning with city braking function and pedestrian warning with city braking function.

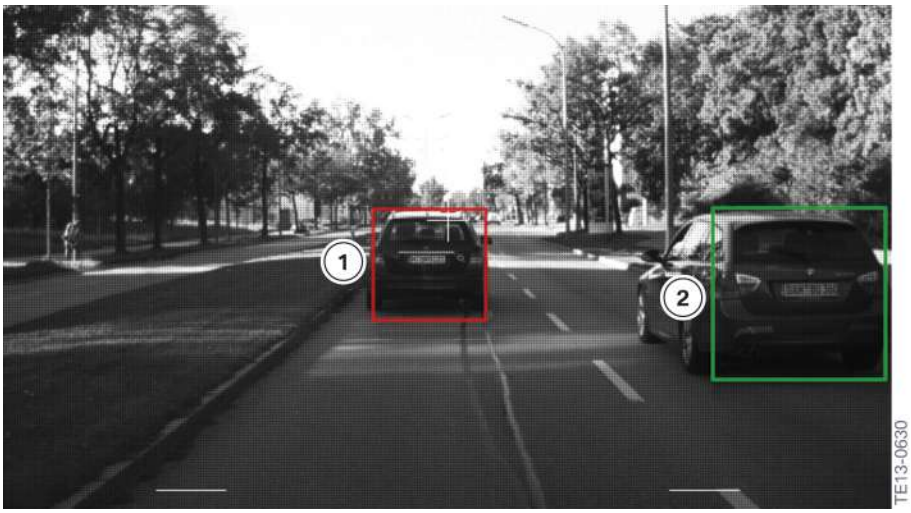


Timeplan for the camera-based collision warning

Index	Explanation
1	Collision warning ("early")
2	Collision warning ("medium")
3	Acute warning (acoustic warning signal, brake system is prepared and brake assistant is adapted)
4	Braking is initiated (city braking function only in the range from 5 km/h to 85 km/h)
5	Detection range of KAFAS camera

Collision warning with city braking function

The collision warning with city braking function is active from approx. 5 km/h to 85 km/h. If there is not reaction by the driver after an acute warning at speeds up to 85 km/h, the vehicle will be braked for 1.5 s in order to avoid a collision.

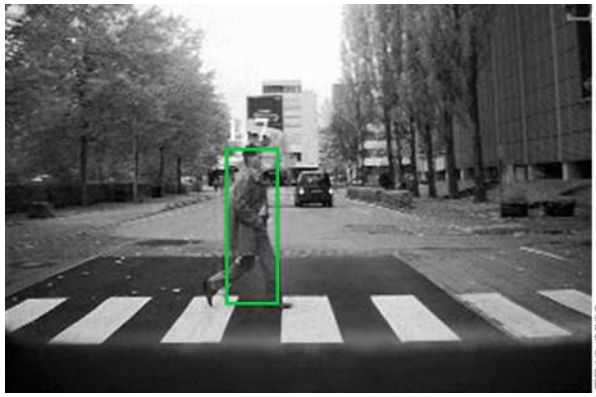


Vehicle identification by KAFAS camera

Index	Explanation
1	Vehicle in same lane
2	Vehicle in different lane

Pedestrian warning with city braking activation function

The system warns the driver about potential collisions with pedestrians in a speed range between 10 km/h and 60 km/h. The vehicle is braked for 1.5 s with the pedestrian warning and city braking function.



Pedestrian detection by KAFAS camera

Warning displays

The following table provides an overview of the different warning displays:

System	Speed range	Early warning	Acute warning	Active brake intervention
Collision warning with city brake function	> 85 km/h	 Early warning from approx. 15 km/h		No. Brake system is prefilled
Collision warning with city brake function	5-85 km/h	—		Yes
Pedestrian warning with city braking activation function	10-60 km/h	—		Yes

4.11. Avoidance assistant

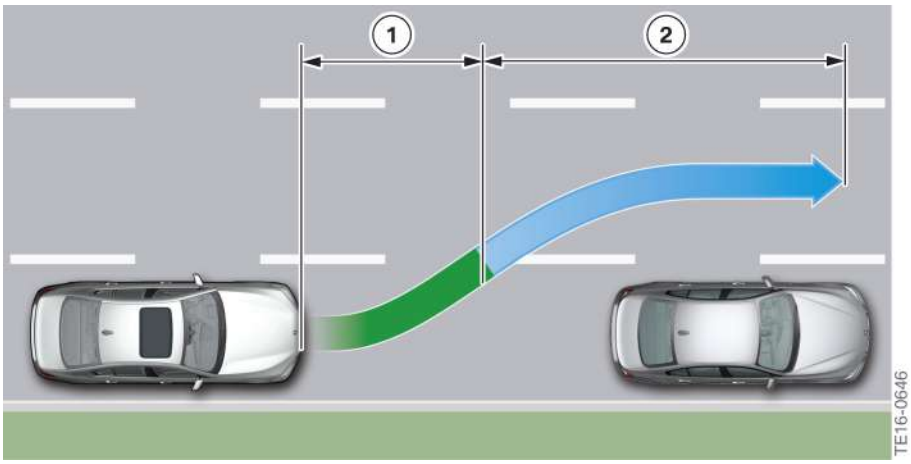
The avoidance assistant supports the driver in critical driving situations with a targeted steering intervention if the driver has to avoid an obstacle. The system also help keep the vehicle stable during and after the avoidance manoeuvre.

The avoidance assistant was used for the first time in the G30.

4.11.1. Function

An emergency avoidance manoeuvre takes place when a suddenly occurring obstacle is detected in front of the vehicle and the driver has to avoid this obstacle quickly and without “significant” braking. The avoidance assistant calculates an optimum “avoidance path” past the obstacle.

The driver initiates the avoidance manoeuvre by turning the steering wheel rapidly, at the same time determining the avoiding direction. Any brake intervention initiated by the acute warning is aborted or suppressed as appropriate. The Electronic Power Steering EPS implements a supporting steering intervention. The vehicle is made more agile by the EPS for avoidance steering and is stabilised again by the Dynamic Stability Control DSC when it is on the avoidance path.

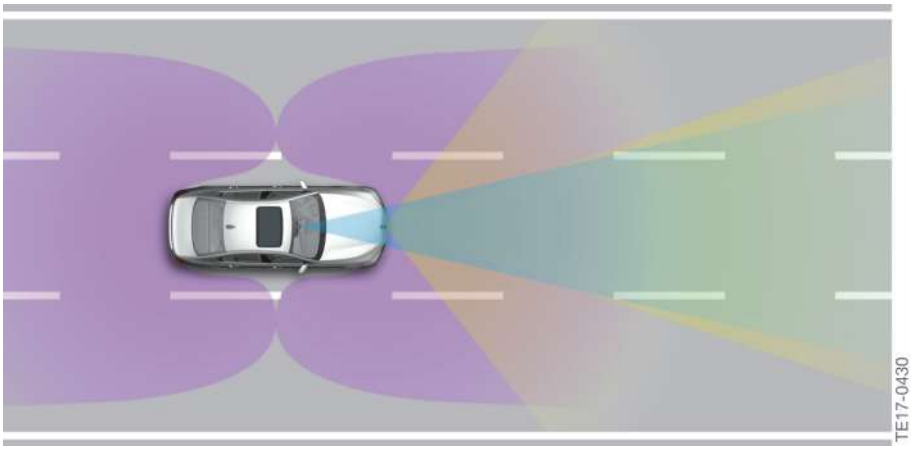


Avoidance assistant

Index	Explanation
1	Make agile
2	Stabilise

4.11.2. Sensors

The radar sensors and KAFAS camera monitor the vehicle surroundings.



Avoidance assistant sensors

4.11.3. Operation

Operation of the avoidance assistant on its own is not possible. The avoidance assistant can be deactivated by switching off all intelligent safety systems (ALL OFF).

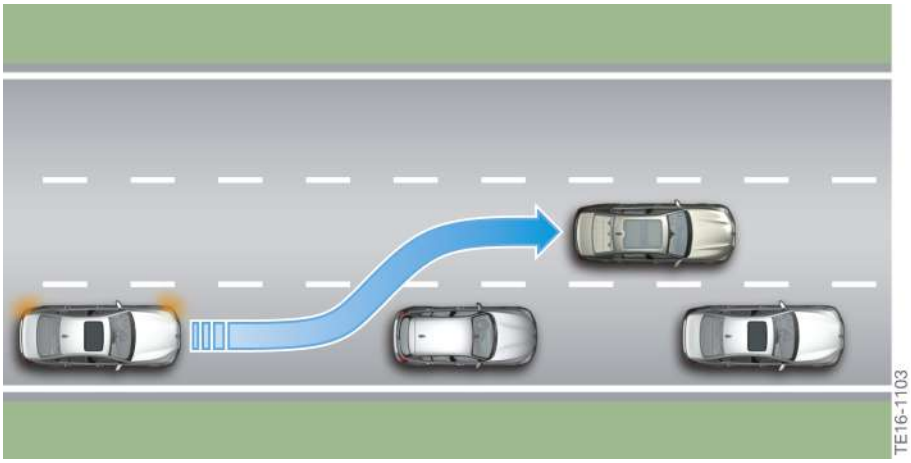
4.12. Lane change assistant

4.12.1. Function

The lane change assistant has been used since the G30.

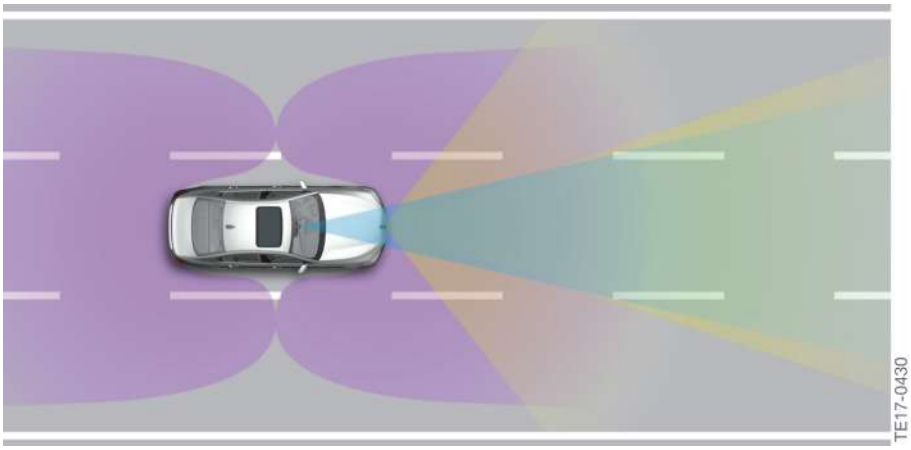
The system assists the driver when changing lanes. If the traffic conditions permit, the vehicle independently performs a steering wheel movement and resumes lane centre guidance again after the lane change. The system assists between 70 km/h and 180 km/h.

The lane change assistant is activated if the driver operates the turn indicator (one-touch signalling) and holds it for at least approximately 1 second) with the steering and lane control assistant function activated. The lane change is performed only if the system does not detect any other road users in a collision-critical range.



Lane change assistant

4.12.2. Sensors



Lane change assistant sensors

4.12.3. Operation

The lane change assistant is active only if the steering and lane control assistant is active. The driver must operate the turn indicator (one-touch signalling) for at least 1 s. Separate operation or configuration is not possible.

4.12.4. Displays

The lane change assistant is displayed in the instrument cluster.

Display	Explanation
	Grey steering wheel symbol: • The system has been interrupted and will not make any further steering wheel movements. The system is in Standby mode. • The system is activated automatically if all conditions are met.
	Steering wheel symbol and lane marking on left flashing green. • Request to change to left lane detected The right side flashes in the case of a request to change lane to the right.

4.13. Crossing traffic warning

4.13.1. Function

The crossing traffic warning function supports the driver when manoeuvring in and out of a parking space and in other situations such as entrances and exits with restricted visibility. The crossing traffic warning function was used for the first time in the G11/G12.

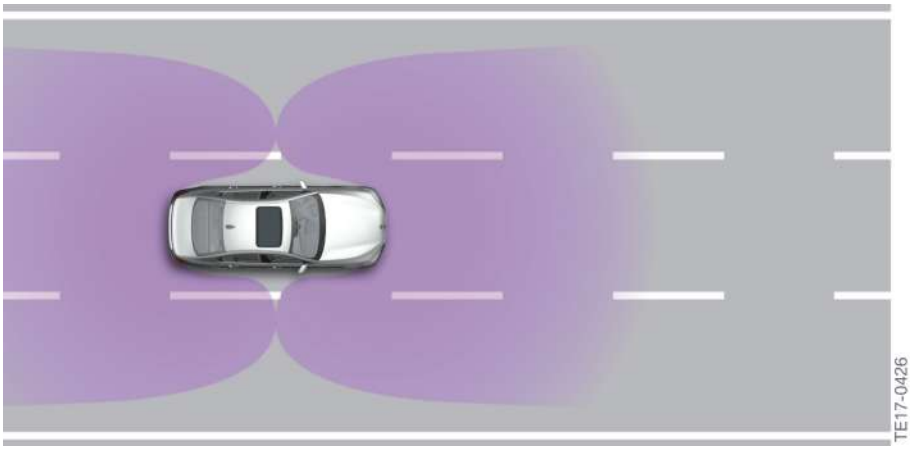
A visual warning and an acoustic warning are output is an object is detected which would be in the area in front of or behind the vehicle within the following 2 s.



Crossing traffic warning

4.13.2. Sensors

The sensors for the crossing traffic warning function are the front and rear radar sensors (not ACC sensor).



Radar sensors for crossing traffic warning

4.13.3. Operation

The crossing traffic warning is switched on and off by means of the controller in the “My vehicle” menu:

The crossing traffic warning is automatically deactivated in the following situations:

- Speed > 7 km/h
- Active steering and lane control assistant
- Active Parking Manoeuvring Assistant.

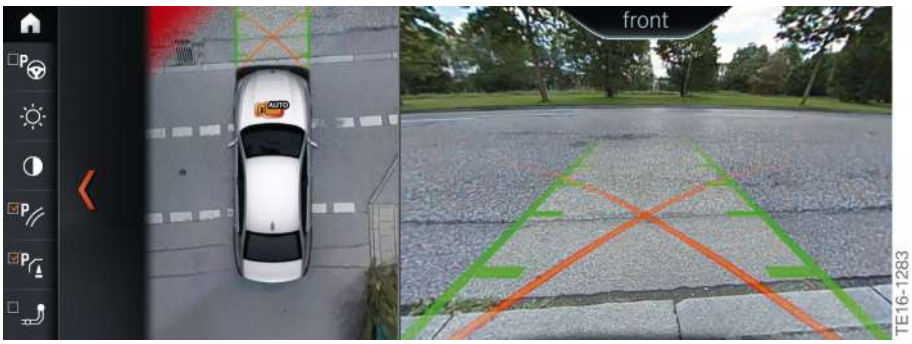
4.13.4. Displays

The visual warning when reversing is output via the lane change warning display in the exterior rearview mirrors.



Display in exterior rearview mirror

The corresponding area flashes red in the PDC view when an object is detected.



Warning in PDC view

The monitored area also flashes red in the camera view if an object is detected.



Warning in camera view

4.14. Junction warning

4.14.1. Function

The junction warning has been offered since the G30.

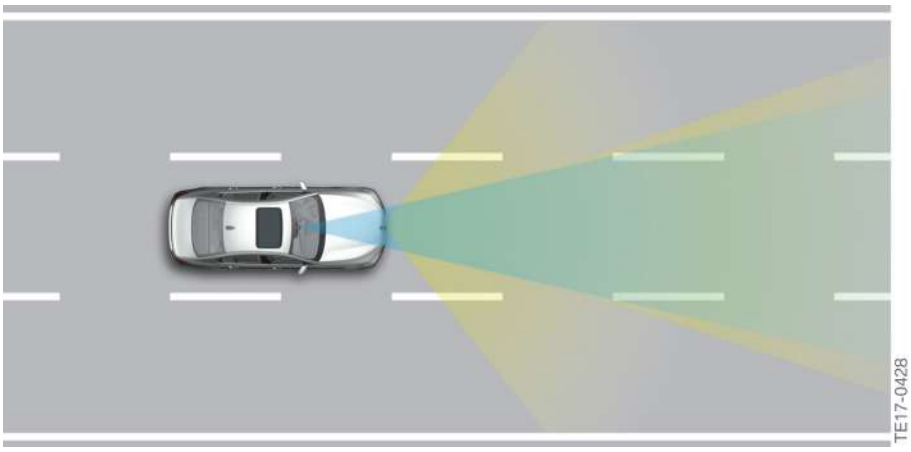
The system warns the driver if there is a risk of collision with crossing traffic. If a danger situation is detected by the junction warning function, it request the driver to perform emergency braking by means of a warning.



Junction warning

4.14.2. Sensors

The KAFAS camera and the ACC sensor are used as sensors for the junction warning function.



Junction warning sensors

4.14.3. Operation

The junction warning function cannot be operated separately. The system can be deactivated only by switching off all intelligent safety systems.

4.14.4. Displays

The visual warning is displayed in the instrument cluster or in the Head-Up Display.



Warning by junction warning function in instrument cluster

4.15. Road sign recognition

4.15.1. Function

The road sign detection function assists the driver in recognising speed limits. The road sign detection function has been used since the F01/F02.

If data on the speed limits for the road on which the vehicle is currently driving is stored in the navigation system, this will be used for the road sign display. Parallel to this, the data of the KAFAS camera is compared with the predefined data. The road signs detected by the KAFAS camera have a higher priority than the data stored in the navigation system. This may be the case when driving through motorway road works, for example.

Certain road signs with additional signs, e.g. "when wet" or pictograms, are also detected. This information is compared with the internal vehicle data, e.g. of the rain sensor, and is displayed correspondingly depending on the situation. The system also takes into account applicable time restrictions which have been identified, e.g. "between 10:00 pm and 6:00 am". Here, the current time of the on-board clock and the time condition stored in the navigation system are used for comparison.

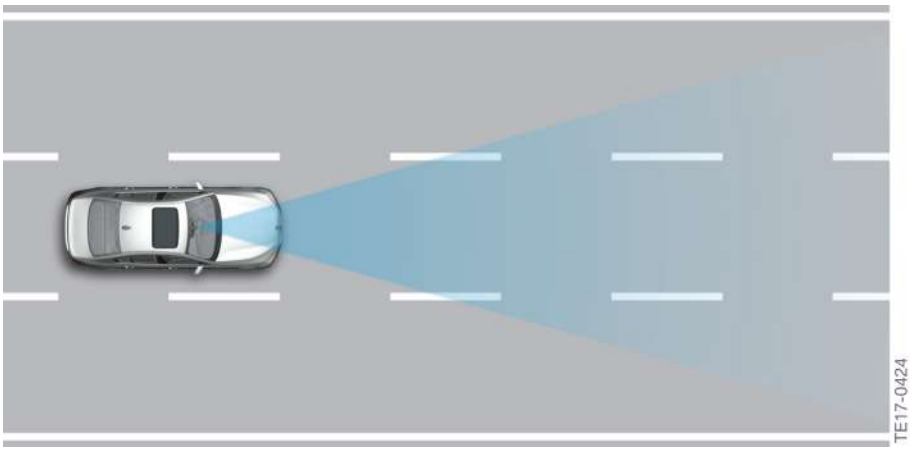
In addition to image data, other information is also used in the evaluation and calculation for the display. This includes:

- Time display (for time-based additional conditions)
- Wiper status (for additional sign "When wet")
- Turn indicators (special speed limits, e.g. at motorway exits)
- Outside temperature sensor for “frost/snow”.

Road signs for speed limits that do not comply with the legal standard, particularly those without circular frames, are not always detected. The same also applies to road signs which are fully or partially covered by stickers, dirt or vegetation.

4.15.2. Sensors

The KAFAS camera is used as the sensor.



Sensor for road sign detection

4.15.3. Operation

The road sign detection function can be activated and deactivated in the “My vehicle” menu.

4.15.4. Displays

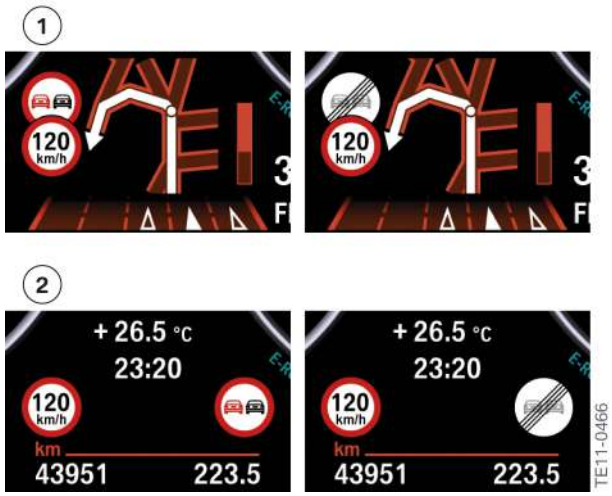
The driver is informed about the speed limits in the instrument cluster or, where applicable, in the Head-Up Display.



Display of road sign detection in the instrument cluster

4.15.5. No-overtaking display

Since the F20, the start and end of no-overtaking zones can also be detected parallel to the speed limits. Detection and display take place in the same way as for the previous road sign detection function.



Display for road sign detection with "no overtaking" sign

Index	Explanation
1	Displays with navigation display in instrument cluster
2	Displays in standard view in the instrument cluster

4.15.6. Road sign recognition without navigation system

For certain models, road sign detection is also available without a navigation system. This currently includes the following models:

- MINI as of F56
- BMW F45/F46.

For the road sign recognition without navigation system only the traffic sign detected by the KAFAS camera is displayed in the instrument cluster. Additional text characters generally cannot be read and interpreted. This variant now allows the road sign detection function to be offered in countries which were previously unable to offer the system due to the lack of navigation data. A quality comparable to a system with navigation is difficult to achieve.

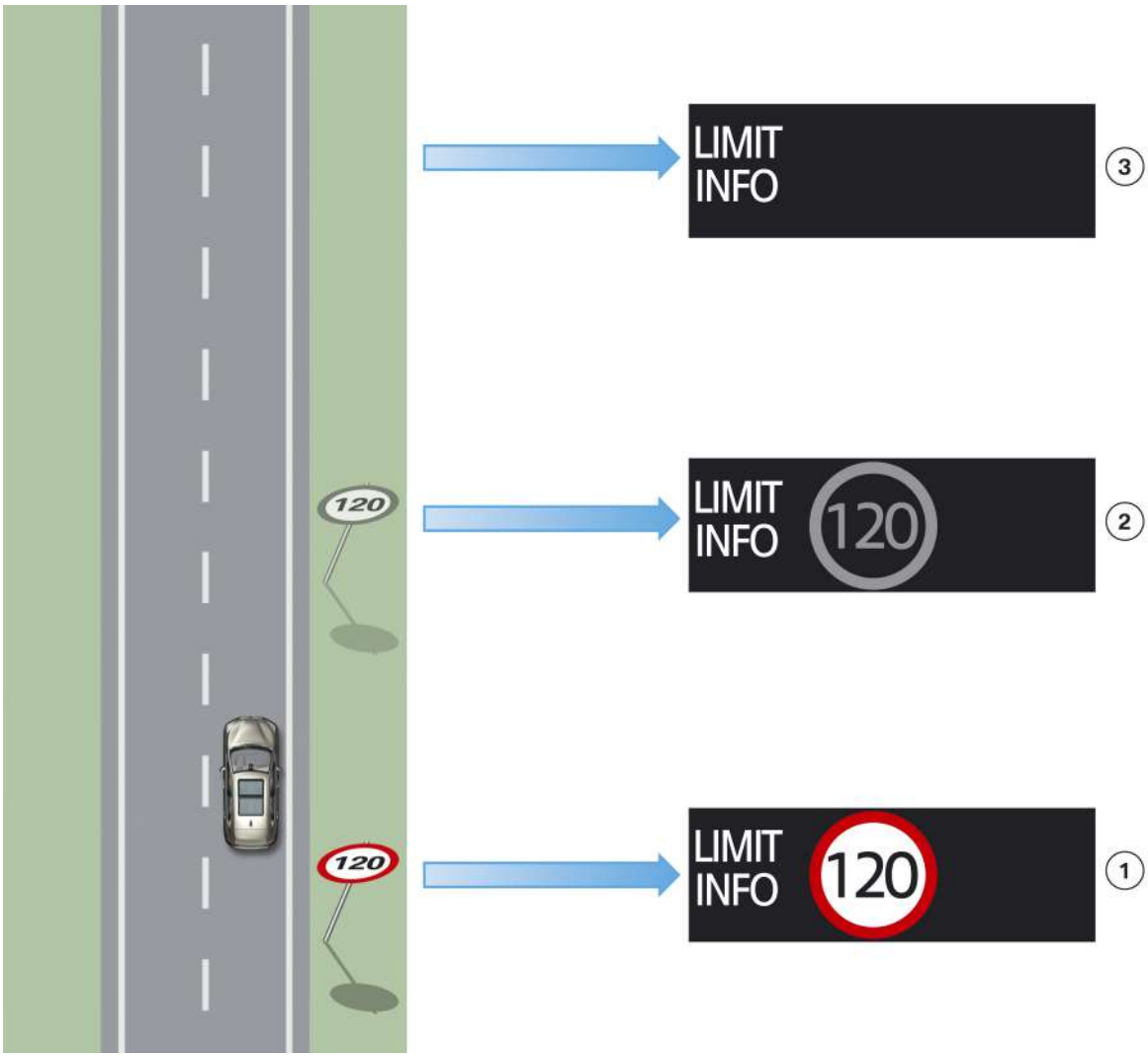
Time restrictions can be recognised as such; however, an evaluation of their validity (time) is not possible without navigation data. Time-based speed limits are therefore initially displayed briefly independently of their validity, and then remain in an "uncertain status" (grey display) for longer.

If the traffic sign is not repeated after a certain distance travelled, the top speed limitation displayed in the instrument cluster is greyed out in order to indicate an "**unsafe** condition" to the driver. After a further distance travelled without the recognition of another road sign, the greyed out speed limit is finally suppressed.

ASSISTANCE SYSTEMS.
4. DRIVING.

The calculation, when a display is set to an unsafe condition (grey display), is dependent on the following factors:

- Respective speed limit
- National-market version of the vehicle
- Internal calculation of the road type on which the vehicle is currently driving.



road sign recognition function without navigation system

Index	Explanation
1	Detection of the traffic sign with display in the instrument cluster
2	• If no other road sign is recognised after a certain distance travelled, the display changes to an **unsafe** condition (grey display). • If another road sign is recognised after a certain distance travelled, the display is retained in the instrument cluster.
3	Symbol is hidden if no road sign has been detected after covering a further distance.

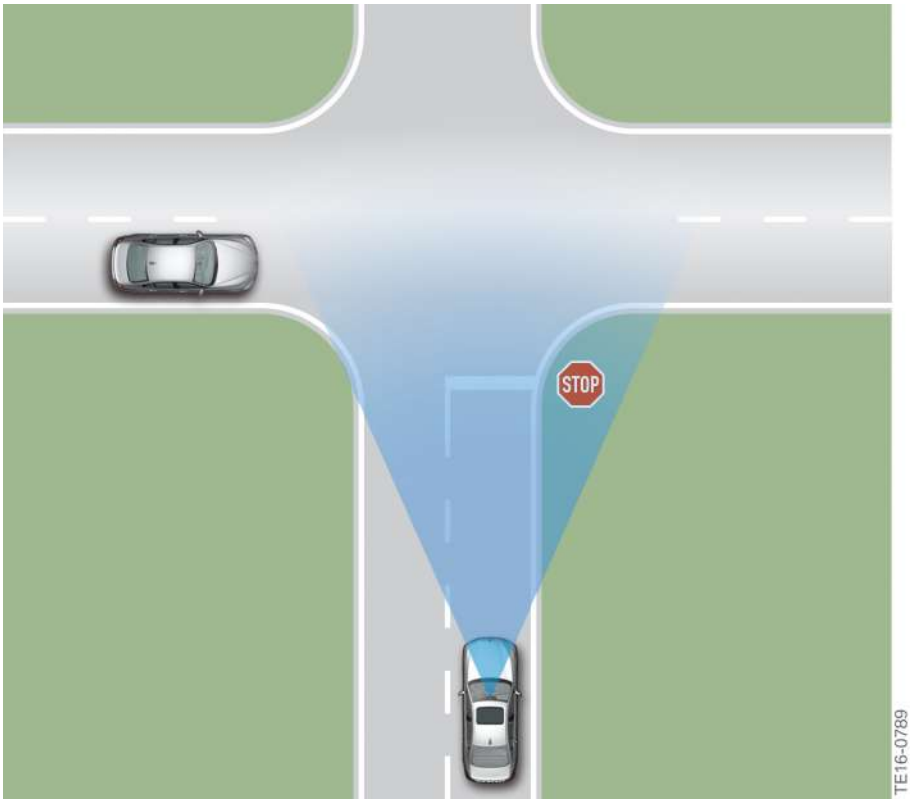
4.16. Junction traffic sign recognition assistant

4.16.1. Function

The junction traffic sign recognition assistant has been offered since the G30.

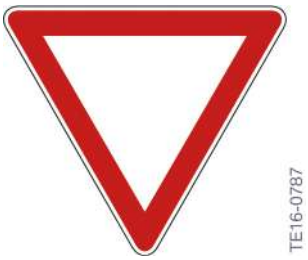
The system warns the driver in situations where the right of way may be ignored. This applies in the following traffic situations, for example:

- Crossroads
- Junction
- Entrance
- Roundabout.



Function of junction traffic sign recognition assistant

The following road signs are detected:



Give Way

An advance warning is issued when a Give Way sign is detected.



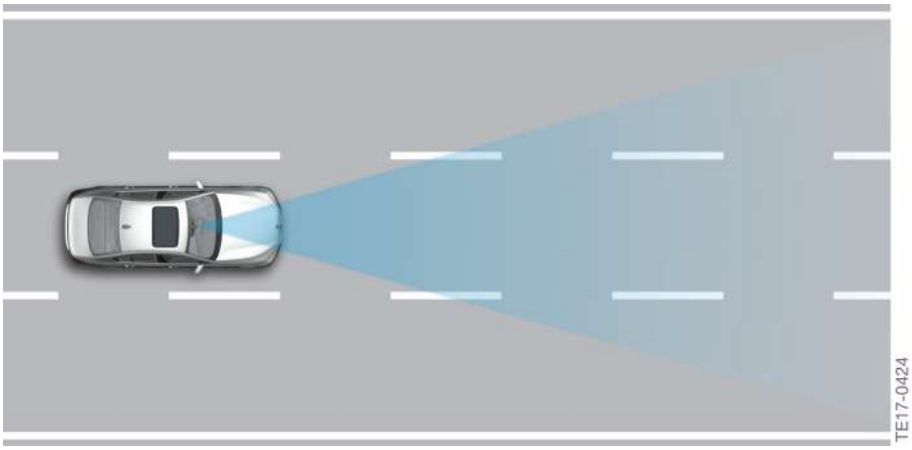
Stop sign

An advance warning and an acute warning are output for a stop sign.

Traffic lights are also detected by the KAFAS camera. No warning is output for a green traffic light and Give Way sign.

4.16.2. Sensors

The KAFAS camera is responsible for detection.



Sensor for junction traffic sign recognition assistant

4.16.3. Operation

The warning threshold values are set in the Intelligent Safety menu.

The warning threshold values affect the assumed braking so that the vehicle arrives at a Stop sign or at a Give Way sign at a speed of approximately 15 km/h (or approximately 35 km/h for a roundabout). These are computed mean values which can vary correspondingly according to the driver's response time. The warning level setting selected by the driver in the Intelligent Safety configuration menu is also an important factor. The "late" warning level, for example, deactivates the visual advance warnings and warns only with an acute warning for Stop signs. A Give Way sign, on the other hand, will be ignored in this setting as there is no acute warning for a Give Way situation. Braking is required in this setting, although this is still far removed from ABS-controlled braking.



Intelligent Safety button

4.16.4. Display

An acoustic warning is also output in addition to the visual warning for the junction traffic sign recognition assistant.



Display for junction traffic sign recognition assistant

4.17. Wrong-way assistant

4.17.1. Function

The wrong-way warning function was offered for the first time in the G30.

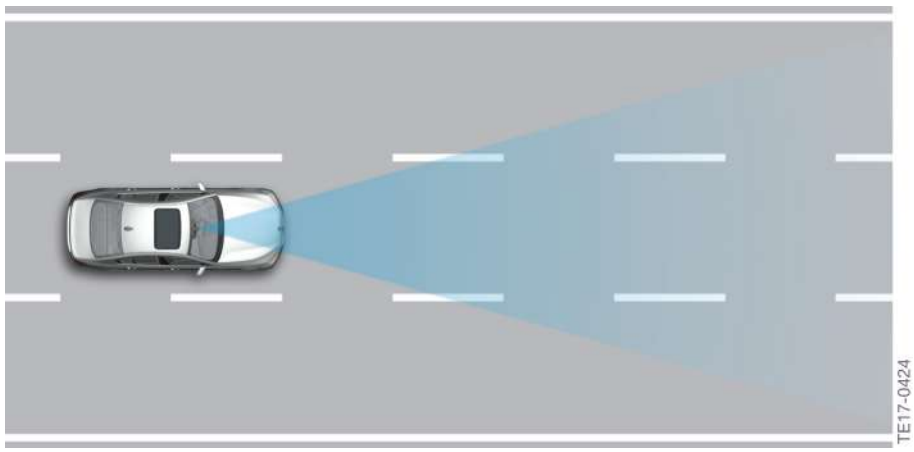
The system helps the driver in confusing traffic situations or road layouts, such as can be found when entering a motorway or a roundabout. The system is also able to recognise one-way streets as such in inner-city zones.

Detection takes place using the KAFAS camera and the navigation data. The following examples are possible:

Recording	Explanation
 TE16-0507	2 prohibition signs + navigation data
 TE16-0509	2 prohibition signs + mandatory passage
 TE16-0510	1 prohibition sign + mandatory passage + navigation data
 TE16-0511	1 roundabout sign + navigation data

4.17.2. Sensors

The KAFAS camera is used as the sensor.


TE17-0424

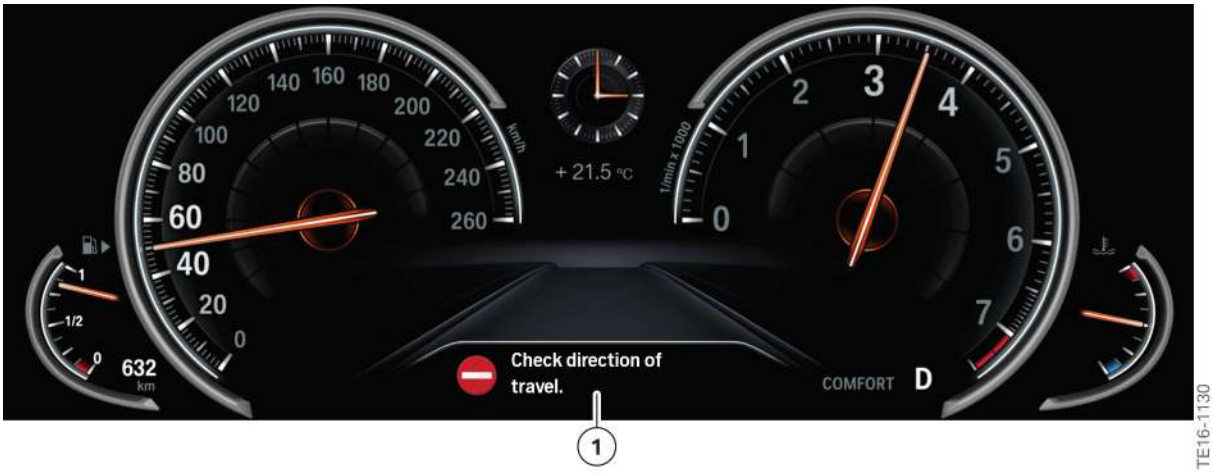
Wrong-way warning sensor

4.17.3. Operation

The wrong-way warning function cannot be switched on and off.

4.17.4. Display

The driver is warned visually and acoustically if he drives the wrong way. The visual warning is output in the instrument cluster and, where applicable, in the Head-Up Display.



Display for wrong-way warning

Index	Explanation
1	Warning: Check direction of travel

4.18. MINI

The MINI models are equipped with the following assistance systems from the "Driving" area:

- Cruise control with braking function
- Speed limit
- Road sign recognition
- Collision warning with city brake function
- Pedestrian warning with city braking activation function
- Active cruise control with stop&go function.

ASSISTANCE SYSTEMS.
5. CHECK CONTROL MESSAGES.

5.1. Displays

The following table provides an overview of the Check Control messages of the assistance systems. An exact description for the current fault message is displayed additionally on the central information display.

Display	Explanation
	Active Cruise Control (ACC)
	Park Distance Control (PDC)
	Speed control
	Night vision
	Camera-based driver support systems (KAFAS)
	High-beam assistant (FLA)
	Lane departure warning

ASSISTANCE SYSTEMS.
5. CHECK CONTROL MESSAGES.

Display	Explanation
	Parking manoeuvring assistant PMA
	Lane change warning SWW
	Tailgate warning
	Traffic Jam Assist
	Pedestrian warning
	Side collision warning
	Speed limit



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